

# ARCANE: Do Role-Playing Language Agents Stay in Character at the Right Time?

Woojung Song · Nalim Kim · Sangjun Song · Chaewon Heo · Jongwon Lim ·  
Yohan Jo

---

## ABSTRACT

*Role-playing language agents (RPLAs) should play characters whose values and behavior evolve as the story progresses, not maintain a fixed persona. Existing benchmarks measure factual recall at a given chapter, not whether responses align with the character’s psychological trajectory, especially in scenarios the source text never explores. We introduce ARCANE (Arc-Aware Narrative Evaluation), an automatically constructed benchmark spanning 17 novels and 80 principal characters. A Character Arc segments the narrative into phases along a psychological axis, and each probe poses the same scenario across phases, spanning both situations within the source text and situations beyond it. Across six models and six context modes, conditioning on the Character Arc tops every other context strategy on every model, and the gap is largest on scenarios outside the source text where retrieval has nothing to find. We further fine-tune open-weight models on the same data to obtain ARCANE-8B/32B, which widen the Arc advantage even more on scenarios outside the source text.*<sup>a</sup>

---

<sup>a</sup>The code and data will be released upon publication.

---

## 1 Introduction

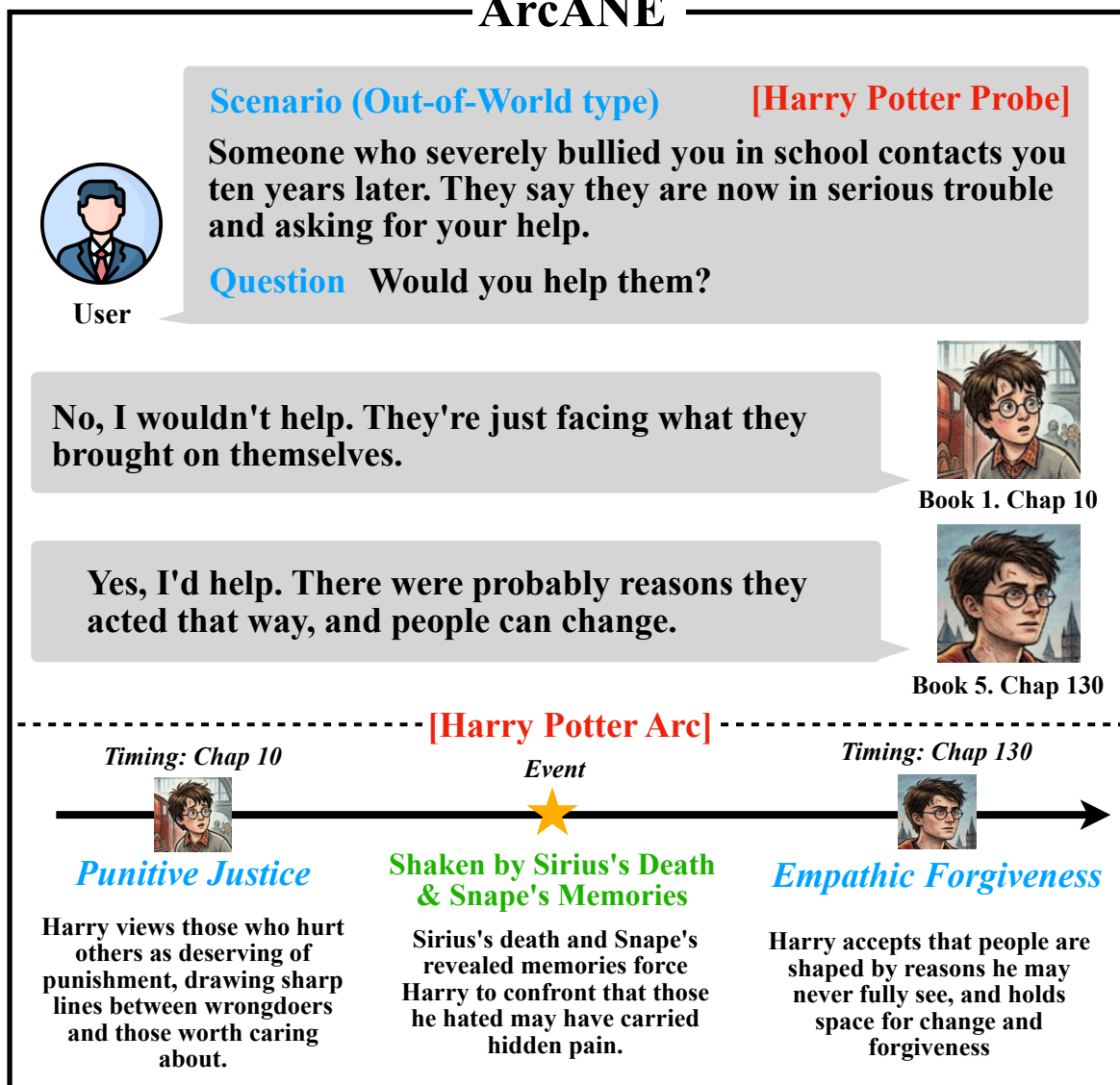
Role-playing language agents (RPLAs) have become one of the most popular applications of conversational AI, enabling users to interact with fictional, historical, or persona-based characters [1, 4]. As these agents have expanded across entertainment, companionship, interactive storytelling [27], and education [21], user expectations have shifted from mere fluency toward authentic character portrayal [32, 39]. RPLAs are thus expected to generate plausible [31], in-character responses that sustain the user’s sense of immersion [41].

---

<sup>\*</sup>Equal contribution.

<sup>†</sup>Corresponding author.

# ArcANE



**FIGURE 1** An example from ARCANÉ: an Out-of-World *probe* elicits different moral responses from Harry across his *Character Arc*, shifting from *Punitive Justice* in Book 1 to *Empathic Forgiveness* in Book 5.

Recent work has framed point-in-time role-playing as a task that situates a character at a specific moment in their timeline, motivated by the need to preserve narrative immersion [2, 28, 29]. While much effort goes into suppressing explicit spoilers [30, 36], this alone is not sufficient. Beyond verifiable facts, a character’s values and behavioral patterns also shift over time—as Figure 1 illustrates, Harry Potter’s moral stance differs sharply between Book 1 and Book 5. Evaluating RPLAs therefore requires assessing whether each response fits the character’s state at that moment.

To address this gap, we aim to evaluate whether an RPLA’s behavior shifts faithfully as a

character evolves across the narrative. We focus on novels as our testbed: unlike short dialogues, scripts, or game transcripts, novels provide extended narratives in which character evolution unfolds vividly across many chapters, offering both rich internal description and explicit temporal structure [40]. The core intuition behind our approach is simple: posing the same question at different points in the story should elicit different responses. This temporal approach goes beyond prior benchmarks such as *TimeCHARA* [2], which queries RPLAs at distinct points in time only to detect *factual* hallucinations (i.e., whether the character knows what they should know at that moment), rather than how they would *behave* in situations beyond their factual knowledge.

**Our Approach.** We instantiate this goal in ARCANÉ (**Arc-Aware Narrative Evaluation**), a benchmark that measures whether an RPLA’s portrayed actions and motivations shift in step with the source character’s evolving personality across narrative phases. To do this, we need (1) a structured way to represent how a character’s psychology changes across the narrative, and (2) a way to query the character at specific points along that change. We address each by introducing *Character Arcs* and *probes*, as illustrated in Figure 1. A *Character Arc* aligns a character’s key events with their evolving psychological states, organizing both into a phase-segmented trajectory along a psychological axis—for instance, Harry’s moral axis moves from *Punitive Justice* (Book 1) to *Empathic Forgiveness* (Book 5) after Sirius’s death and Snape’s revealed memories. Each *probe*, built on top of an arc, consists of a scenario, a question, and reference actions and thoughts that specify how the character should respond at each phase.

Our dataset covers 17 novels and 80 principal characters, providing 544 character arcs and 4,601 probes. In designing probes, we consider that users of RPLAs are often less interested in what already happens in the source text than in how the character would behave in entirely new situations they themselves might encounter. Probes are therefore constructed not only from source-grounded scenarios but also from situations the source text never explores. The dataset is released in two splits: a training set that supplies supervision for arc-grounded responses, and an evaluation set whose probes are constructed only after every underlying character arc has been reviewed and updated through consensus by three independent human annotators.

We use ARCANÉ for two complementary perspectives of evaluation. The first, our main experimental focus, is the *context-strategy* perspective: which form of narrative context, supplied to an RPLA at inference time, most effectively grounds the model in the character’s state at the queried phase. The second is the *model* perspective: which LLMs exhibit less character-behavioral hallucination at chapter-anchored probes. We further include ARCANÉ-8B and ARCANÉ-32B, trained on our training set, in the same comparison.

Across model families and sizes, Arc-grounded context achieves the best performance, and this advantage extends to low-popularity titles outside the validated slice. Surprisingly, Arc’s advantage becomes larger when the probe scenario is not in the source text. In these cases, retrieval-based approaches have nothing relevant to find, while Arc still provides the character’s state at that point. The same pattern carries over to other role-playing model families, and fine-tuning on our training set further enhances performance, particularly in situations beyond the source text.

| Benchmark          | Auto-constructed? | Free-form? | In-narrative? | Out-of-narrative? | Point-in-time? | Temporal behavioral? | Evaluated via        |
|--------------------|-------------------|------------|---------------|-------------------|----------------|----------------------|----------------------|
| PsychoBench [35]   | ✗                 | ✗          | ✗             | ✗                 | ✗              | ✗                    | Trait                |
| InCharacter [39]   | ✓                 | ✓          | ✗             | ✗                 | ✗              | ✗                    | Trait                |
| CharacterBox [37]  | ✓                 | ✓          | ✓             | ✓                 | Δ              | ✗                    | Simulation           |
| CoSER [40]         | ✓                 | ✓          | ✓             | ✗                 | Δ              | ✗                    | Simulation           |
| Character-LLM [32] | ✓                 | ✓          | ✗             | ✓                 | ✗              | ✗                    | Behavior             |
| CharacterEval [36] | ✓                 | ✓          | ✓             | ✗                 | Δ              | ✗                    | Behavior             |
| RoleBench [38]     | ✓                 | ✓          | ✗             | ✓                 | ✗              | ✗                    | Behavior             |
| ChatHaruhi [16]    | ✓                 | ✓          | ✓             | ✗                 | ✗              | ✗                    | Behavior             |
| LifeChoice [43]    | ✓                 | ✗          | ✓             | ✗                 | ✓              | ✗                    | Behavior             |
| HPD [5]            | ✓                 | ✓          | ✓             | ✗                 | ✓              | ✓                    | Behavior             |
| RoleEval [33]      | ✗                 | ✗          | ✓             | ✗                 | ✗              | ✗                    | Knowledge-Based Only |
| TimeCHARA [2]      | ✓                 | ✓          | ✓             | ✗                 | ✓              | ✗                    | Knowledge-Based Only |
| ArcANE(Ours)       | ✓                 | ✓          | ✓             | ✓                 | ✓              | ✓                    | Behavior             |

**TABLE 1** Comparison of role-playing language agent evaluation benchmarks. **Auto-constructed**: data generated primarily by LLMs. **Free-form**: open-ended response generation. **In-narrative**: scenario-based evaluation within source text. **Out-of-narrative**: scenarios beyond the source text. **Point-in-time**: evaluation anchored to specific narrative timepoints. **Temporal behavioral**: assesses behavioral changes across narrative timepoints. **Evaluated via**: modality through which persona is assessed. Δ: partial support.

## 2 Related Work

### 2.1 Role-Playing Language Agents

Role-playing language agents (RPLAs), in which LLMs simulate specific characters, have been deployed across diverse applications including interactive fiction, game NPCs, emotional companions, and personalized assistants [4, 6, 10, 31, 38, 42]. Beyond such practical utility, cognitive psychology research has shown that character simulation serves as an abstraction of human psychology and social experience, allowing people to explore minds, relationships, and possible selves through fictional figures [20, 25].

What users seek from RPLAs is consequently not mere response generation but an *immersive experience of a living character*: an agent whose behavior remains coherent and responsive as the interaction unfolds, sustaining the user’s narrative engagement [3, 11, 28, 29] with the figure being portrayed. A widely reported failure mode is that agents gradually lose persona grounding and produce generic responses even when the context remains intact [19, 34]. Faithfully sustaining such a living character is therefore the central capability against which RPLA techniques must be evaluated.

### 2.2 Benchmarking Role-Playing Language Agents

Benchmarks for RPLAs have developed around different targets of measurement: the character’s trait inventory [35, 39], factual knowledge and persona-driven decisions [2, 32, 43], surface representation such as linguistic style and conversational competence [16, 36, 38, 45], and behavioral consistency across turns [36, 37]. Across these dimensions, however, the character is treated as a *static target*—an identity to be reproduced rather than one whose behavior shifts as events accumulate [13, 15, 17].

Moving beyond this assumption, simulation-based benchmarks [37, 40] place characters in

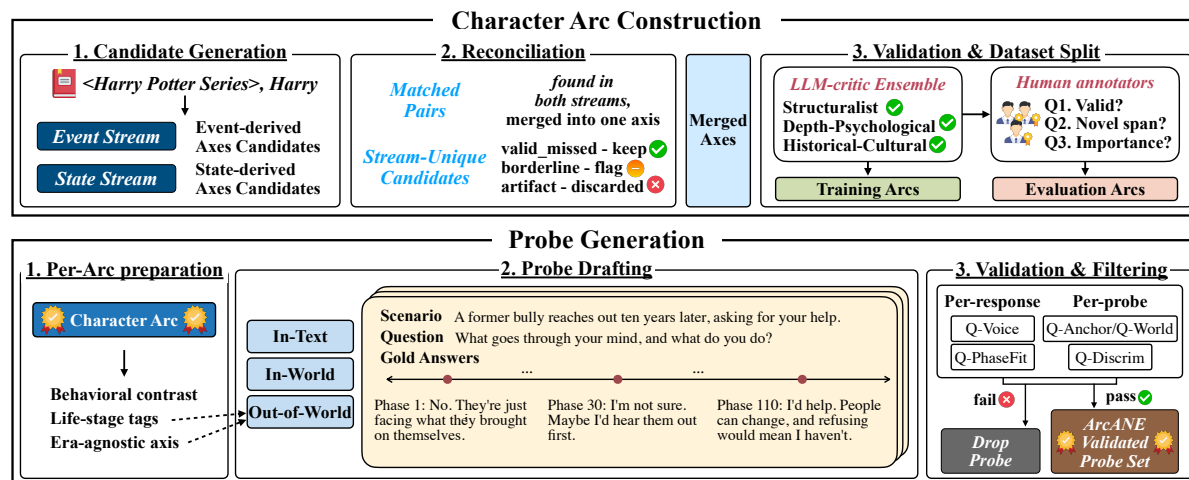


FIGURE2 The ARCANE pipeline: constructing character arcs (top) and evaluation probes (bottom).

multi-turn scenarios to elicit dynamic behavior, and event-conditioned representations [18, 26] encode persona as snapshots or behavior chains. However, these capture variation from inter-character interaction or generic behavior chains rather than from a single character’s events accumulating along the narrative. Benchmarks that consider narrative timepoints (Table 1) remain limited to closed-form decisions [43] or factual recall [2]—a gap we address by probing a character’s behavior at specific narrative timepoints.

### 3 ARCANE

ARCANE (**Arc-Aware Narrative Evaluation**) measures whether an RPLA’s responses shift with the character’s state at the specific phase, rather than matching a fixed persona. Prior RPLA benchmarks largely evaluate a character’s trait inventory [35, 39], which corresponds to McAdams’ Layer 1 [22, 23]: a stable set of dispositions the character carries throughout the story. We move to Layer 2: whether the model expresses those traits at the right moment.

The construction proceeds in two stages. First, *Character Arc Construction* (§3.1) extracts what changes about each character, building one *Character Arc* for every *axis* the character moves along. An axis is a single psychological dimension defined by two pole descriptions. Figure 1 shows one such arc, Harry’s moral axis running from *Punitive Justice* to *Empathic Forgiveness* across the *phases*. Second, *Probe Generation* (§3.2) tests whether a model expresses the right phase at the right moment. Each arc yields probes consisting of a scenario-question pair with one phase-specific reference per phase, so that asking the same scenario at every phase distinguishes a model that tracks the arc from one that recites a fixed persona. Figure 1 also shows an Out-of-World probe built on the same Harry arc, where the scenario “a former bully reaches out ten years later, asking for your help” draws a refusal at an early phase, hesitation in the middle, and acceptance at a late phase.

### 3.1 Character Arc Construction

A Character Arc (Figure 1) aligns a character’s key events with their evolving psychological states, organizing both into a phase-segmented trajectory from an initial state to a final state. Each phase specifies a chapter range, a description of the character’s current state, and the key moments that anchor that state. The three stages below correspond to the top panel of Figure 2.

**Stage 1: Candidate Generation.** The novel is processed through two independent chapter-level streams: an *event stream* extracts psychologically impactful events, and a *state stream* emits a cross-sectional psychological profile. The split keeps event omission separable from state misreading even when both streams see the same chapters. Each stream then induces two types of candidate axes: *intrapersonal* axes track internal change (beliefs, motives, coping), and *relational* axes track a dyadic relation (trust, esteem, intimacy, antagonism). Every candidate is grounded in established literary or psychological scholarship [14] (Figure 6).

**Stage 2: Reconciliation.** An analyst LLM reconciles the candidate axes produced by the event and state streams in Stage 1. Matched pairs—axes proposed by both streams—are merged into a single axis with a direction label that records which pole each phase sits closer to. Each remaining unmatched candidate, proposed by only one stream, is then classified as a genuine missed axis (kept), an ambiguous case (flagged), or a stream-specific artifact (discarded).

**Stage 3: Validation and Dataset split.** The previous stage establishes internal reliability within a single LLM, not external validity. We therefore pass every axis through an critic LLM ensemble of three literary perspectives. For training novels, an axis is retained if at least two of the three critics judge it literarily grounded. For evaluation novels, the critic ratings serve only as reference: three human annotators independently re-assess each axis, and only those judged valid by at least two annotators enter the evaluation set.

### 3.2 Probe Generation

A probe asks how a character would respond to one scenario at different phases of their arc. A single (scenario, question) is paired with  $N$  reference responses, one per phase. Because the same scenario is asked at every phase, success requires more than knowing the character’s overall personality. The model must identify which phase the character is in and respond from there. Each scenario falls into one of three categories that vary by distance from the source text. *In-Scenario* lifts the scene from a verbatim passage, *In-World* invents an unwritten situation inside the source’s setting, and *Out-of-World* transposes the scenario to a non-source era. These three categories form a difficulty gradient. In-Scenario can be answered from the source passage, In-World from the source’s setting, and Out-of-World only from the arc itself. Each response pairs an action with a 1–2 sentence *thought* that captures the cognitive construal when overt action is constrained [24], plus a knowledge-cutoff chapter that blocks reference to later events. A final validation pass discards probes whose scenarios violate their category’s rules or whose per-phase responses end up too similar to tell apart. The three stages below correspond to the bottom panel of Figure 2.



**Stage 1: Per-arc preparation.** Before drafting any probe, we extract three reusable elements per arc. The *behavioral contrast* pins down the axis as a single yes/no decision whose answer differs across phases, giving the designer the axis in concrete decisional form. A per-phase *life-stage tag* from {child, adolescent, young\_adult, adult, older\_adult} specifies the character’s life stage at that phase. For Out-of-World only, an *era-agnostic axis* specifies which era the scenario is transposed to and how the axis manifests in that setting.

**Stage 2: Probe drafting.** For each (target phase, category) pair, a designer LLM drafts one probe with  $N$  phase responses. The target phase’s response reflects how the character actually behaves at that phase, while the other  $N-1$  are counterfactual responses that project each remaining phase’s behavior onto the same scenario, grounded in the arc so that each phase’s projection reads as a plausible response. Figure 2 illustrates this with a concrete example.

**Stage 3: Validation and filtering.** Each drafted probe is validated in two passes. The response pass applies two checks per response. Q-Voice verifies that the response stays in character, free of anachronism, and respects the knowledge cutoff. Q-PhaseFit asks a blind LLM judge which phase the response best fits, and flags any mismatch with the target phase. A failure on either check triggers one regeneration, and if it persists, the response is marked **unavailable** while the rest of the probe is kept. The probe pass applies two checks per probe. Q-Anchor (In-Scenario) or Q-World (In-World, Out-of-World) verifies that the scenario respects the setting’s rules, and a failure drops the entire probe. Q-Discrim flags adjacent phase pairs whose responses are not separated along the behavioral contrast, but this check is annotation-only, since some overlap between adjacent phases is theoretically expected [9].

### 3.3 ARCANE Dataset

ARCANE covers 17 novels, 80 principal characters, 544 arcs, and 4,601 probes, partitioned into three disjoint subsets (Table 3): a *training* slice (10 novels, 2,545 probes), from which 45,690 teacher-generated SFT rows are pooled; a *validated evaluation* slice (5 novels, 1,754 probes), the headline benchmark with every axis cleared by the Stage iii critic ensemble and a 2-of-3 human-validity majority; and an *unvalidated low-popularity* slice (2 novels, 302 probes), held out from human annotation and drawn from the bottom of the Project Gutenberg popularity distribution as a memorization control (§4.4). Details are in Appendix C.

### 3.4 Training Arc-aware RPLAs

Although ARCANE is introduced primarily as an evaluation framework, its structure lends itself to a training formulation. Because each probe is answered across multiple phases of a character’s arc, the dataset naturally yields contrastive pairs that could teach a model to distinguish subtle behavioral shifts between adjacent developmental stages.

To verify that this formulation translates into measurable training gains, we fine-tune Qwen3-8B and Qwen3-32B [44] through a two-stage pipeline. In the SFT stage, the model learns the response format of ARCANE. In the subsequent DPO stage, it learns to distinguish ground-truth character behavior from plausible but temporally displaced alternatives, namely actions or speech

that fit the character but belong to a different narrative phase. Full data construction and training details are provided in Appendix D.

## 4 Experiments

| Family      | Size  | Mode       | In-Scenario |      |      |      | In-world |      |      |      | Out-of-world |      |      |      | Overall |
|-------------|-------|------------|-------------|------|------|------|----------|------|------|------|--------------|------|------|------|---------|
|             |       |            | APF         | RPF  | RAE  | PTF  | APF      | RPF  | RAE  | PTF  | APF          | RPF  | RAE  | PTF  |         |
| DeepSeek-V4 | Flash | Vanilla    | 59.8        | 59.3 | 52.1 | 53.8 | 56.7     | 56.9 | 48.8 | 45.4 | 54.8         | 55.4 | 46.8 | 44.5 | 52.9    |
|             |       | Summary    | 64.1        | 63.0 | 56.4 | 59.6 | 57.8     | 58.1 | 49.4 | 49.4 | 57.2         | 57.6 | 49.0 | 47.7 | 55.8    |
|             |       | RAG        | 66.3        | 64.7 | 59.1 | 60.9 | 56.6     | 57.1 | 48.0 | 47.8 | 54.0         | 54.8 | 45.2 | 44.2 | 54.9    |
|             |       | LifeChoice | 66.3        | 64.7 | 59.1 | 61.7 | 57.4     | 57.8 | 48.6 | 49.5 | 56.8         | 57.0 | 48.2 | 46.5 | 56.1    |
|             |       | TimeChara  | 60.8        | 60.2 | 53.2 | 55.6 | 56.2     | 56.6 | 48.4 | 46.8 | 54.5         | 55.2 | 46.3 | 43.5 | 53.1    |
|             |       | Arc        | 66.8        | 65.8 | 59.2 | 62.1 | 61.8     | 61.9 | 53.8 | 55.3 | 62.2         | 62.0 | 54.5 | 51.4 | 59.7    |
|             | Pro   | Vanilla    | 61.3        | 60.2 | 53.4 | 56.1 | 54.2     | 54.9 | 46.1 | 45.5 | 53.4         | 54.1 | 45.2 | 44.0 | 52.4    |
|             |       | Summary    | 67.1        | 65.6 | 60.2 | 61.4 | 59.7     | 59.5 | 51.8 | 51.2 | 57.1         | 57.4 | 48.8 | 46.5 | 57.2    |
|             |       | RAG        | 68.1        | 66.1 | 61.8 | 63.0 | 57.6     | 57.6 | 49.2 | 50.5 | 55.7         | 56.1 | 47.6 | 47.0 | 56.7    |
|             |       | LifeChoice | 69.1        | 66.9 | 63.0 | 64.0 | 59.1     | 58.9 | 51.1 | 50.9 | 56.7         | 56.8 | 48.7 | 47.6 | 57.7    |
|             |       | TimeChara  | 60.4        | 59.5 | 53.1 | 54.8 | 54.9     | 55.4 | 46.9 | 46.7 | 53.0         | 53.7 | 44.8 | 42.5 | 52.1    |
|             |       | Arc        | 69.5        | 67.7 | 62.8 | 65.1 | 64.2     | 63.6 | 57.0 | 58.2 | 64.1         | 63.7 | 57.0 | 55.8 | 62.4    |
| Qwen3       | 8B    | Vanilla    | 43.0        | 42.8 | 32.7 | 38.2 | 40.5     | 41.0 | 30.8 | 30.6 | 41.4         | 42.4 | 32.1 | 30.6 | 37.2    |
|             |       | Summary    | 47.2        | 45.4 | 36.5 | 41.7 | 42.1     | 41.9 | 32.1 | 31.6 | 41.7         | 41.9 | 31.7 | 30.3 | 38.7    |
|             |       | RAG        | 53.0        | 51.0 | 43.2 | 44.6 | 43.0     | 43.1 | 33.1 | 31.6 | 42.2         | 42.7 | 32.8 | 30.3 | 40.9    |
|             |       | LifeChoice | 50.9        | 48.9 | 40.8 | 42.4 | 42.6     | 42.3 | 32.1 | 31.4 | 42.3         | 42.2 | 32.1 | 30.0 | 39.8    |
|             |       | TimeChara  | 43.1        | 43.1 | 33.3 | 36.5 | 41.6     | 42.2 | 32.0 | 30.6 | 41.8         | 42.8 | 32.7 | 29.4 | 37.4    |
|             |       | Arc        | 50.1        | 48.5 | 39.2 | 42.1 | 47.5     | 47.3 | 37.3 | 36.6 | 47.4         | 47.5 | 37.4 | 36.0 | 43.1    |
|             | 32B   | Vanilla    | 49.7        | 48.6 | 39.4 | 43.4 | 48.6     | 48.6 | 38.8 | 35.8 | 48.6         | 48.6 | 39.2 | 35.0 | 43.7    |
|             |       | Summary    | 54.8        | 53.6 | 45.2 | 46.7 | 49.5     | 49.3 | 39.2 | 36.0 | 48.5         | 48.7 | 38.9 | 36.6 | 45.6    |
|             |       | RAG        | 59.4        | 57.3 | 50.8 | 49.2 | 49.6     | 49.9 | 40.0 | 38.6 | 48.3         | 48.7 | 39.0 | 36.1 | 47.2    |
|             |       | LifeChoice | 58.3        | 56.6 | 49.1 | 50.5 | 50.6     | 50.5 | 40.9 | 40.3 | 48.4         | 48.6 | 38.8 | 36.3 | 47.4    |
|             |       | TimeChara  | 51.3        | 50.4 | 41.3 | 45.4 | 48.9     | 49.1 | 39.5 | 36.4 | 48.4         | 48.9 | 39.3 | 34.4 | 44.4    |
|             |       | Arc        | 57.4        | 55.6 | 47.4 | 51.6 | 54.1     | 53.7 | 44.3 | 41.0 | 54.8         | 54.3 | 45.4 | 41.7 | 50.1    |
| ARCANE      | 8B    | Vanilla    | 47.8        | 46.9 | 38.3 | 41.9 | 48.2     | 47.5 | 39.0 | 37.4 | 47.4         | 46.8 | 38.5 | 36.3 | 43.0    |
|             |       | Summary    | 56.5        | 55.0 | 47.7 | 49.7 | 49.4     | 48.8 | 40.0 | 40.3 | 50.3         | 49.6 | 41.2 | 38.3 | 47.2    |
|             |       | RAG        | 61.4        | 59.5 | 54.4 | 53.0 | 48.8     | 48.8 | 40.0 | 39.6 | 49.6         | 49.4 | 41.2 | 36.3 | 48.5    |
|             |       | LifeChoice | 58.2        | 56.2 | 50.1 | 50.5 | 49.1     | 48.6 | 40.1 | 39.3 | 50.3         | 49.9 | 41.7 | 37.9 | 47.6    |
|             |       | TimeChara  | 49.3        | 48.2 | 40.2 | 43.0 | 47.8     | 47.3 | 38.8 | 38.9 | 47.7         | 47.4 | 39.3 | 36.5 | 43.7    |
|             |       | Arc        | 59.5        | 57.7 | 50.6 | 56.0 | 60.9     | 59.3 | 52.3 | 54.6 | 62.5         | 60.5 | 54.5 | 54.6 | 56.9    |
|             | 32B   | Vanilla    | 53.6        | 53.2 | 44.9 | 45.0 | 55.1     | 55.1 | 46.7 | 41.7 | 56.5         | 56.2 | 48.6 | 41.5 | 49.9    |
|             |       | Summary    | 57.0        | 55.8 | 48.4 | 49.5 | 53.2     | 52.6 | 44.0 | 44.3 | 51.9         | 51.6 | 42.6 | 40.6 | 49.3    |
|             |       | RAG        | 65.8        | 63.5 | 59.2 | 55.8 | 52.9     | 52.9 | 43.8 | 41.9 | 51.6         | 51.8 | 42.9 | 41.6 | 52.0    |
|             |       | LifeChoice | 62.1        | 60.2 | 54.6 | 53.6 | 54.9     | 54.2 | 45.8 | 44.3 | 54.4         | 53.9 | 45.3 | 41.1 | 52.0    |
|             |       | TimeChara  | 53.7        | 53.2 | 44.8 | 46.9 | 55.6     | 55.0 | 47.5 | 41.8 | 56.4         | 56.1 | 48.7 | 39.6 | 49.9    |
|             |       | Arc        | 59.4        | 58.0 | 50.9 | 53.4 | 66.2     | 64.2 | 58.8 | 58.5 | 68.0         | 65.7 | 61.6 | 59.8 | 60.4    |

**TABLE 2** Main results on the validated slice (5 novels). Probe categories and metrics are defined in §3.2 and §4.2. Best mode per column per model is bolded.

### 4.1 Experimental Setup

We evaluate six models under six in-context strategies on the validated evaluation slice of §3.3: five novels (*Harry Potter*, *Anna Karenina*, *Don Quixote*, *The Count of Monte Cristo*, and *The Autobiography of Benjamin Franklin*) covering 25 principal characters, 205 arcs, and 1,754 probes.

**Models.** We compare four open-weight baselines, Qwen3-8B, Qwen3-32B [44], DeepSeek-V4-Flash, and DeepSeek-V4-Pro [7], and our post-trained models, ARCANE-8B and ARCANE-32B.

**Context modes.** *Vanilla* supplies only the character’s identity and the query chapter. *Summary* adds the most recent five chapter summaries. *RAG* adds the top-6 retrieved source-text chunks both truncated at the query chapter. *LifeChoice* [43] and *TimeCHARA* [2] reuse the corresponding



original context formats. *ARC* (ours) supplies the curated character arc truncated at the phase of the query chapter, exposing the model to the same trajectory information used to construct the probes. Full context-mode details are in Appendix E; all prompts are in Appendix P.

## 4.2 Evaluation Protocol

The role-playing language agent receives the probe (scenario, question, and the chapter-truncated context produced by the current mode) and emits a free-form response. An LLM judge (DeepSeek-V4-Flash; full setup, prompts, and human-annotator plausibility validation in Appendix F) scores the response against the reference attached to the probe phase on a 1–100 scale, at two granularities.

**Per-phase metrics** Three sub-scores per (probe, phase) call. **APF (Action Phase-Fidelity)** grades the response’s overt action against `ref_action` at three levels (strategy, valence, target). **RPF (Reasoning Phase-Fidelity)** parses both `ref_thought` and the response’s reasoning into four mechanism slots (trigger, appraisal, goal, strategy) and aggregates the slot-wise matches. **RAE (Reasoning–Action Entailment)** fixes `ref_thought` as the operative reasoning and asks whether the response’s action is one this reasoning would license, catching responses that are internally coherent yet reach an action `ref_thought` would not.

**Trajectory metric** Per-phase metrics do not penalise a model that scores well on each phase in isolation yet collapses adjacent phases or moves along the axis in the wrong direction. **PTF (Phase Trajectory Fidelity)** addresses this: a single judge call sees all  $N$  phase-keyed (reference, response) pairs together and returns the mean of three sub-scores (alignment, direction, shape).

## 4.3 Main Results

Across the six models and six context modes from §4.1, conditioning on the Character Arc gives the strongest Overall score on every model. Table 2 reports Overall and the four per-category metrics for each (model, mode) cell. The Arc row leads the Overall column for all six models, with a gap of 2.2 to 8.4 points over the same model’s strongest non-Arc row. At the high-capacity end, DeepSeek-V4-Pro reaches 62.4 under Arc against 57.7 for its best non-Arc mode (LifeChoice). At the small open-weight end, Qwen3-8B reaches 43.1 under Arc against 40.9 for its best non-Arc mode (RAG). The pattern survives a per-novel split: across the 30 (model, novel) Overall cells produced by evaluating each of the six models on each of the five validated novels, Arc is the top mode on 29 (per-novel breakdown in Appendix L).

The lift is not uniform across the three probe categories. The Arc-vs-best-non-Arc gap on Overall in the DeepSeek-V4-Pro row is +0.5 on In-Scenario, +5.2 on In-World, and +7.7 on Out-of-World. The same ordering recurs in the other models, and the probe construction explains it. An In-Scenario probe lifts its scenario from a verbatim source passage, so source-text retrieval (RAG, LifeChoice) already returns the relevant scene and Arc adds little. In-World and Out-of-World probes have no such passage, and only Arc supplies the phase of the query chapter.

The Arc lift is consistently larger on the trajectory metric (PTF) than on the per-phase metrics (APF, RPF, RAE). APF, RPF, and RAE judge each (probe, phase) pair in isolation, so

a non-Arc mode matching plausible per-phase content can score well phase by phase without shifting along the arc. PTF grades all  $N$  phase responses to a probe as one sequence (§4.2) and rewards only sequences whose alignment, direction, and shape track the reference across phases. For DeepSeek-V4-Pro, the Arc-vs-best-non-Arc PTF gap reaches +1.1 on In-Scenario, +7.0 on In-World, and +8.2 on Out-of-World, each at least as large as the matched gap averaged over the three per-phase metrics (+0.3, +4.6, +7.2). The same ordering holds for the other models. Arc is the only mode that supplies the per-chapter trajectory PTF scores against.

## 4.4 Additional Results

We extend §4.3 in two directions: low-popularity novels outside the validated subset, and role-playing models beyond the Table 2. The Arc lift carries over to two low-popularity titles. We rerun the six models on *The Underdogs* (469 downloads/month) and *East Lynne* (1,038/month), each at least two times below the lowest-download validated title, using the same probe pipeline.<sup>1</sup> Arc is again the top Overall mode for all six models, with a lift of +4.1 to +15.3 over the strongest non-Arc mode (Table 6 in Appendix I). The largest lift is on ARCANE-32B (+15.3), whose Out-of-World Arc average on these two novels (70.6) is 6.8 points above the validated novel value (63.8). Character-class results are in Appendix M.

Figure 3 carries the §4.3 Arc-lift pattern to other role-playing models and isolates the SFT and DPO contributions inside ARCANE. It plots the per-category Arc gap to best baseline for HER-32B [8], CoSER-8B and CoSER-70B [40], and the four ARCANE model at both sizes (full numbers in Appendix J). Every entry has a non-negative lift on In-World and Out-of-World, and In-Scenario stays mixed: the asymmetry generalizes beyond Table 2. Inside ARCANE, SFT broadly raises Arc Overall from 50.1 to 58.4 but its Arc lift on In-World and Out-of-World averages only +7.3 at ARCANE-32B-SFT. DPO reaches further into those categories, growing it to +12.5 at ARCANE-32B-DPO, at the cost of a smaller In-Scenario lift from −2.2 to −5.6. We analyze this pattern in detail in §5.3.

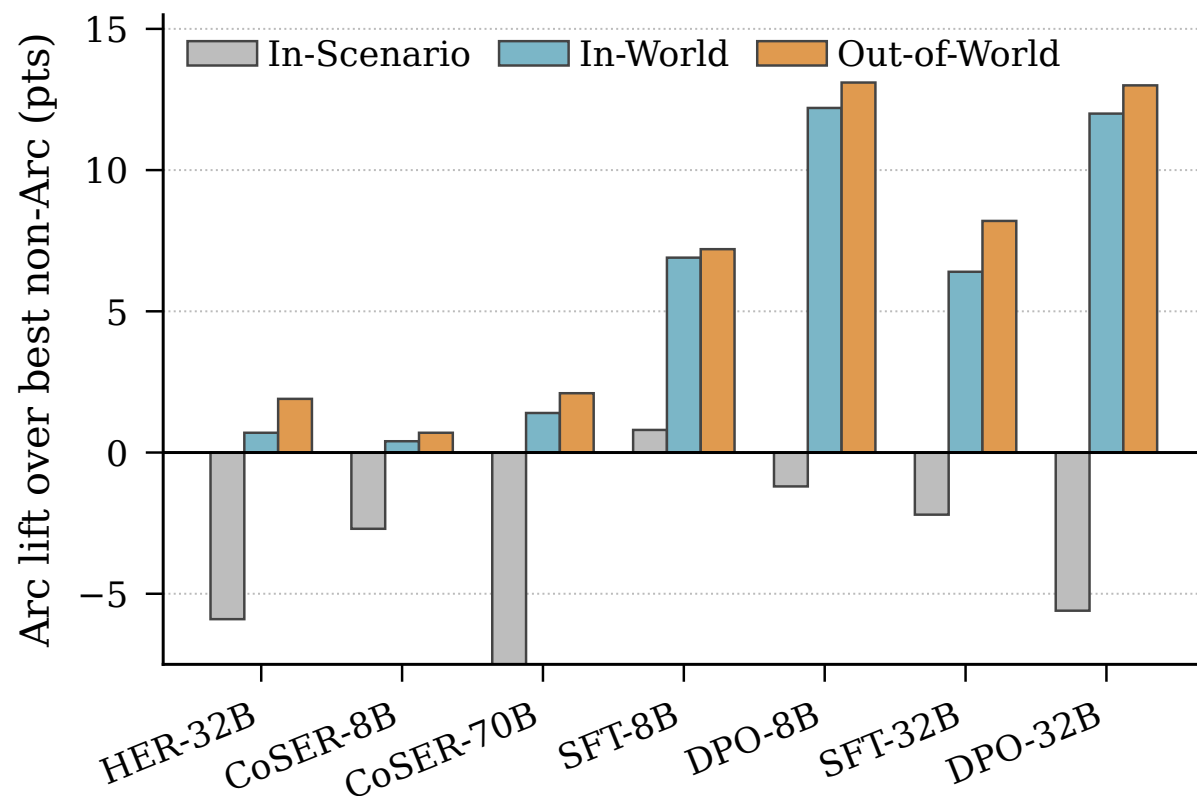
# 5 Analysis

## 5.1 Source-of-effect ablation

Two ablations test whether the Arc lift in Table 2 reflects the per-phase content rather than a structured-context bonus or a leakage of the probes’ generative source. MIXEDARC swaps in another character’s arc from the same novel. ARCHINT strips the per-phase prose down to one **Axis** / **Phase** line per axis, an approximately 40× context compression. We run both on approximately 120 validated probes per model for DeepSeek-V4-Flash, Qwen3-32B, and ARCANE-32B-DPO (see Figure 4). Construction details and the probe sample are in Appendix K.

MIXEDARC fails to clear Vanilla on the two models (Qwen3-32B and ARCANE-32B-DPO) where Arc itself produces a large gain, and on ARCANE-32B-DPO it falls −4.7 Per-Phase

<sup>1</sup>Validated public-domain titles, downloads/month: Monte Cristo 80,951, Anna Karenina 17,785, Don Quixote 12,669, Benjamin Franklin 2,187. Harry Potter is non-Gutenberg.



**FIGURE 3** Per-category Arc lift (Arc minus best non-Arc) for added RPLA baselines and the ARCANE models. SFT/DPO at each size are adjacent. Tables 2 and 7.

Average and  $-6.5$  PTF *below* Vanilla: the wrong content actively misleads a trained model, so neither the structured-context-bonus nor the source-leakage account survives. ARCHINT then localises where the inference-time effect lives. On DeepSeek-V4-Flash and Qwen3-32B it tracks Arc within  $\pm 2.6$  points on every metric, so the axis label paired with the current-phase index (both pipeline outputs from §3.1) already carries the full Arc advantage for prompting. On ARCANE-32B-DPO it recovers only half the gap ( $-5.1$  Per-Phase Average), attributing the  $+10.3$  DPO gain in §4.3 to per-phase-prose conditioning rather than the structural index. The pipeline therefore yields two products the ablation cleanly separates: a label-and-index hint sufficient for prompting general-purpose models, and a per-phase prose substrate the training step requires.

## 5.2 Evaluation validation

**Human-anchored judge plausibility.** Three annotators rate the DeepSeek-V4-Flash judge verdict on 70 probes drawn across 7 novels and all 6 context modes. Majority plausibility is 87.1%, with all three categories falling within 10 points (In-Scenario 84.6%, In-World 82.3%, Out-of-World 95.7%). On a 50-cell subset the annotators also re-scored APF/RPF/RAE on the same 1–100 scale, anchored to the judge’s verdict; the human-average Overall tracks the judge

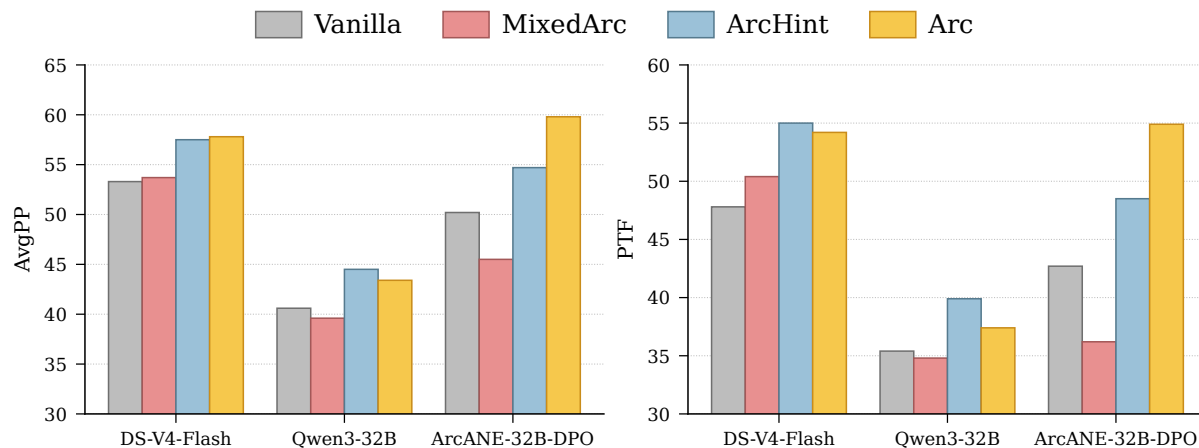


FIGURE 4 Arc source-of-effect ablation across three models, validated subset. DS refers to DeepSeek.

at Pearson  $r=0.96$ , so where humans adjusted the judge they shifted it by only a few points. Per-dimension correlations and the full re-score table are in Appendix O.1.

**LLM cross-judge replication.** On an independent 300-cell sample stratified across 5 novels and 4 (model, mode) configs, we re-score with three different judges (Claude Sonnet 4.5, Opus 4.5, and GPT-5.5). DeepSeek-V4-Flash sits inside the three-judge consensus, and all four judges rank ARCANE-32B-DPO under Arc first, and the full ranking table are in Appendix O.2.

**PTF metric sensitivity.** Perturbing the probe response sequences (response-shuffle, response-reverse, and block-shuffle, with per-dimension drops reported in Appendix N) drops PTF Overall on ARCANE-32B-DPO by  $-8.8$  to  $-23.0$  points against  $-1.3$  to  $-6.3$  on DeepSeek-V4-Pro: PTF responds to phase-order corruption, and the larger drop on the model with stronger per-phase anchoring reads as a behavioural signature of phase tracking rather than a per-phase re-summary.

### 5.3 Training-effect analysis

ARCANE-32B-DPO beats Qwen3-32B on 1,198/1,750 validated subset probes (ties 84, losses 468, mean Overall  $\Delta=+9.49$ ), with the category asymmetry of §4.3. Side-by-side inspection of the wins surfaces one stylistic difference between the two models: ARCANE writes in the first-person, present-tense register the judges prefer, while Qwen3-32B defaults to third-person narration. The hostile reading: the gain is a register artifact, not arc tracking. We test this with a POV control. On a stratified 150-probe sample we append one register instruction to Qwen3-32B’s prompt (“respond IN CHARACTER, first person, present tense”) and rescore the regenerated responses under the unchanged Stage-iii rubric. Forcing the register *lowers* Qwen’s Overall to 50.0 (from 53.8), while ARCANE holds at 56.7. If register carried the gain, the instruction would close it. A proposition-level decomposition on the same 150 probes attributes the surviving gap to category-conditional canon discipline, not surface register (Appendix G, Table 5).

**What does DPO add to SFT?** DPO beats SFT on 940/1,750 probes, with the same category trade. SFT wins In-Scenario, DPO widens on In-World and Out-of-World. The step concentrates on trajectory direction. SFT often holds one static voice across phases. DPO moves with the arc. On Don Quixote’s interdependence axis, SFT keeps the same knightly tag in every phase, while DPO drops it in the repentant phase and turns toward family. The one character where the first-person register ARCANE installs is a net loss, Hagrid, whose communal idiom sits more naturally in third-person scene narration, is documented in Appendix H.

## 6 Conclusion

We ask whether role-playing language agents track the right character state at the right point in a narrative. We introduce ARCANE, an automated benchmark of 544 *Character Arcs* and 4,601 probes across 17 novels. Across six models and six context modes, Arc-grounded context tops Overall on every model, and the gap widens from In-Scenario to Out-of-World where retrieval has nothing to find. The lift carries to two held-out low-popularity novels (+4.1 to +15.3 over the best non-Arc mode), and our ARCANE-32B models extend it further. Multi-turn dialogue, where the arc itself advances through interaction, is the natural next step.

## Limitations

While ARCANE offers a new perspective for evaluating role-playing language agents, several limitations remain. Our dataset is constructed entirely in English and restricted to the novel domain, since our evaluation relies on extended narratives in which character evolution unfolds across many chapters. Our evaluation also centers on a single character’s behavior over accumulating events, and does not capture user-character or character-character interactions. Higher-fidelity role-playing agents may also enable more credible impersonation, and downstream applications should surface character-AI status and avoid presenting generations as authoritative interpretations. Finally, our dataset draws on 19th- and early-20th-century novels whose period-bound social attitudes may be reproduced by trained models, so released artifacts are intended for research only.

## References

- [1] Character AI. <https://character.ai/>. Accessed: 2026-04-26.
- [2] Jaewoo Ahn, Taehyun Lee, Junyoung Lim, Jin-Hwa Kim, Sangdoo Yun, Hwaran Lee, and Gunhee Kim. TimeChara: Evaluating point-in-time character hallucination of role-playing large language models. In Lun-Wei Ku, Andre Martins, and Vivek Srikumar, editors, *Findings of the Association for Computational Linguistics: ACL 2024*, pages 3291–3325, Bangkok, Thailand, August 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.findings-acl.197. URL <https://aclanthology.org/2024.findings-acl.197/>.

- [3] Rick Busselle and Helena Bilandzic. Measuring narrative engagement. *Media Psychology*, 12(4):321–347, 2009. doi: 10.1080/15213260903287259. URL <https://doi.org/10.1080/15213260903287259>.
- [4] Jiangjie Chen, Xintao Wang, Rui Xu, Siyu Yuan, Yikai Zhang, Wei Shi, Jian Xie, Shuang Li, Ruihan Yang, Tinghui Zhu, Aili Chen, Nianqi Li, Lida Chen, Caiyu Hu, Siye Wu, Scott Ren, Ziquan Fu, and Yanghua Xiao. From persona to personalization: A survey on role-playing language agents. *Transactions on Machine Learning Research*, 2024. ISSN 2835-8856. URL <https://openreview.net/forum?id=xr070E8UIZ>. Survey Certification.
- [5] Nuo Chen, Yan Wang, Haiyun Jiang, Deng Cai, Yuhan Li, Ziyang Chen, Longyue Wang, and Jia Li. Large language models meet harry potter: A dataset for aligning dialogue agents with characters. In Houda Bouamor, Juan Pino, and Kalika Bali, editors, *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 8506–8520, Singapore, December 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.findings-emnlp.570. URL <https://aclanthology.org/2023.findings-emnlp.570/>.
- [6] Nuo Chen, Yan Wang, Yang Deng, and Jia Li. The oscars of ai theater: A survey on role-playing with language models, 2025. URL <https://arxiv.org/abs/2407.11484>.
- [7] DeepSeek-AI. Deepseek-v4: Towards highly efficient million-token context intelligence, 2026.
- [8] Chengyu Du, Xintao Wang, Aili Chen, Weiyuan Li, Rui Xu, Junteng Liu, Zishan Huang, Rong Tian, Zijun Sun, Yuhao Li, Liheng Feng, Deming Ding, Pengyu Zhao, and Yanghua Xiao. Her: Human-like reasoning and reinforcement learning for llm role-playing, 2026. URL <https://arxiv.org/abs/2601.21459>.
- [9] William Fleeson. Toward a structure- and process-integrated view of personality: traits as density distribution of states. *Journal of personality and social psychology*, 80 6:1011–27, 2001. URL <https://api.semanticscholar.org/CorpusID:13805210>.
- [10] Jingsheng Gao, Yixin Lian, Ziyi Zhou, Yuzhuo Fu, and Baoyuan Wang. LiveChat: A large-scale personalized dialogue dataset automatically constructed from live streaming. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 15387–15405, Toronto, Canada, July 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.acl-long.858. URL <https://aclanthology.org/2023.acl-long.858>.
- [11] Melanie Green and Timothy Brock. The role of transportation in the persuasiveness of public narratives. *Journal of Personality and Social Psychology*, 79:701–721, 11 2000. doi: 10.1037/0022-3514.79.5.701.
- [12] Jongwook Han, Woojung Song, Jonggeun Lee, and Yohan Jo. Quantifying data contamination in psychometric evaluations of LLMs. In Vera Demberg, Kentaro Inui, and Lluís Marquez, editors, *Findings of the Association for Computational Linguistics: EACL 2026*, pages 6070–6088, Rabat, Morocco, March 2026. Association for Computational



- Linguistics. ISBN 979-8-89176-386-9. doi: 10.18653/v1/2026.findings-eacl.319. URL <https://aclanthology.org/2026.findings-eacl.319/>.
- [13] Pengrui Han, Rafal Dariusz Kocielnik, Peiyang Song, Ramit Debnath, Dean Mobbs, Anima Anandkumar, and R. Michael Alvarez. The personality illusion: Revealing dissociation between self-reports & behavior in LLMs, 2026. URL <https://openreview.net/forum?id=FRXNMF0to7>.
  - [14] Saffron Huang, Esin DURMUS, Kunal Handa, Miles McCain, Alex Tamkin, Michael Stern, Jerry Hong, and Deep Ganguli. Values in the wild: Discovering and mapping values in real-world language model interactions. In *Second Conference on Language Modeling*, 2025. URL <https://openreview.net/forum?id=zJHZJC1G1Z>.
  - [15] Seungbeen Lee, Seungwon Lim, Seungju Han, Giyeong Oh, Hyungjoo Chae, Jiwan Chung, Minju Kim, Beong-woo Kwak, Yeonsoo Lee, Dongha Lee, Jinyoung Yeo, and Youngjae Yu. Do LLMs have distinct and consistent personality? TRAIT: Personality testset designed for LLMs with psychometrics. In Luis Chiruzzo, Alan Ritter, and Lu Wang, editors, *Findings of the Association for Computational Linguistics: NAACL 2025*, pages 8412–8452, Albuquerque, New Mexico, April 2025. Association for Computational Linguistics. ISBN 979-8-89176-195-7. doi: 10.18653/v1/2025.findings-naacl.469. URL <https://aclanthology.org/2025.findings-naacl.469/>.
  - [16] Cheng Li, Ziang Leng, Chenxi Yan, Junyi Shen, Hao Wang, Weishi MI, Yaying Fei, Xiaoyang Feng, Song Yan, HaoSheng Wang, Linkang Zhan, Yaokai Jia, Pingyu Wu, and Haozhen Sun. Chatharuhi: Reviving anime character in reality via large language model, 08 2023.
  - [17] Jingkai Li. Personality traits in large language models: A psychometric evaluation, 2025. URL <https://openreview.net/forum?id=im8tT8Hpfu>.
  - [18] Rui Li, Heming Xia, Xinfeng Yuan, Qingxiu Dong, Lei Sha, Wenjie Li, and Zhifang Sui. How far are LLMs from being our digital twins? a benchmark for persona-based behavior chain simulation. In Wanxiang Che, Joyce Nabende, Ekaterina Shutova, and Mohammad Taher Pilehvar, editors, *Findings of the Association for Computational Linguistics: ACL 2025*, pages 15738–15763, Vienna, Austria, July 2025. Association for Computational Linguistics. ISBN 979-8-89176-256-5. doi: 10.18653/v1/2025.findings-acl.813. URL <https://aclanthology.org/2025.findings-acl.813/>.
  - [19] Pedro Henrique Luz de Araujo, Michael A. Hedderich, Ali Modarressi, Hinrich Schuetze, and Benjamin Roth. Persistent personas? role-playing, instruction following, and safety in extended interactions. In Vera Demberg, Kentaro Inui, and Lluís Marquez, editors, *Proceedings of the 19th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 5329–5359, Rabat, Morocco, March 2026. Association for Computational Linguistics. ISBN 979-8-89176-380-7. doi: 10.18653/v1/2026.eacl-long.246. URL <https://aclanthology.org/2026.eacl-long.246/>.

- [20] Raymond A. Mar and Keith Oatley. The function of fiction is the abstraction and simulation of social experience. *Perspectives on Psychological Science*, 3(3):173–192, 2008. doi: 10.1111/j.1745-6924.2008.00073.x. URL <https://doi.org/10.1111/j.1745-6924.2008.00073.x>. PMID: 26158934.
- [21] Julia M. Markel, Steven G. Opferman, James A. Landay, and Chris Piech. Gpteach: Interactive training with gpt-based students. In *Proceedings of the Tenth ACM Conference on Learning @ Scale, L@S '23*, page 226–236, New York, NY, USA, 2023. Association for Computing Machinery. ISBN 9798400700255. doi: 10.1145/3573051.3593393. URL <https://doi.org/10.1145/3573051.3593393>.
- [22] Dan P. McAdams. What do we know when we know a person? *Journal of Personality*, 63(3):365–396, 1995. doi: <https://doi.org/10.1111/j.1467-6494.1995.tb00500.x>. URL <https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1467-6494.1995.tb00500.x>.
- [23] Dan P. McAdams. The psychological self as actor, agent, and author. *Perspectives on Psychological Science*, 8(3):272–295, 2013. doi: 10.1177/1745691612464657. URL <https://doi.org/10.1177/1745691612464657>. PMID: 26172971.
- [24] Walter Mischel and Yuichi Shoda. A cognitive-affective system theory of personality: reconceptualizing situations, dispositions, dynamics, and invariance in personality structure. *Psychological review*, 102 2:246–68, 1995. URL <https://api.semanticscholar.org/CorpusID:5944664>.
- [25] Keith Oatley. Fiction: Simulation of social worlds. *Trends in Cognitive Sciences*, 20(8): 618–628, 2016. ISSN 1364-6613. doi: <https://doi.org/10.1016/j.tics.2016.06.002>. URL <https://www.sciencedirect.com/science/article/pii/S1364661316300705>.
- [26] Jeyoon Park, Chanjun Park, and Heuseok Lim. CharacterGPT: A persona reconstruction framework for role-playing agents. In Weizhu Chen, Yi Yang, Mohammad Kachuee, and Xue-Yong Fu, editors, *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 3: Industry Track)*, pages 287–303, Albuquerque, New Mexico, April 2025. Association for Computational Linguistics. ISBN 979-8-89176-194-0. doi: 10.18653/v1/2025.naacl-industry.24. URL <https://aclanthology.org/2025.naacl-industry.24/>.
- [27] Joon Sung Park, Joseph O’Brien, Carrie Jun Cai, Meredith Ringel Morris, Percy Liang, and Michael S. Bernstein. Generative agents: Interactive simulacra of human behavior. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology, UIST '23*, New York, NY, USA, 2023. Association for Computing Machinery. ISBN 9798400701320. doi: 10.1145/3586183.3606763. URL <https://doi.org/10.1145/3586183.3606763>.
- [28] Marie-Laure Ryan. *Narrative as virtual reality: Immersion and interactivity in literature and electronic media*. The Johns Hopkins University Press., 2003.

- [29] Marie-Laure Ryan. Interactive narrative, plot types, and interpersonal relations. In *ICIDS*, 2008. doi: [https://doi.org/10.1007/978-3-540-89454-4\\_2](https://doi.org/10.1007/978-3-540-89454-4_2).
- [30] Nafis Sadeq, Zhouhang Xie, Byungkyu Kang, Prarit Lamba, Xiang Gao, and Julian McAuley. Mitigating hallucination in fictional character role-play. In Yaser Al-Onaizan, Mohit Bansal, and Yun-Nung Chen, editors, *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 14467–14479, Miami, Florida, USA, November 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.findings-emnlp.846. URL <https://aclanthology.org/2024.findings-emnlp.846/>.
- [31] Murray Shanahan, Kyle McDonell, and Laria Reynolds. Role-play with large language models, 2023. URL <https://arxiv.org/abs/2305.16367>.
- [32] Yunfan Shao, Linyang Li, Junqi Dai, and Xipeng Qiu. Character-LLM: A trainable agent for role-playing. In Houda Bouamor, Juan Pino, and Kalika Bali, editors, *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 13153–13187, Singapore, December 2023. Association for Computational Linguistics. URL <https://aclanthology.org/2023.emnlp-main.814/>.
- [33] Tianhao Shen, Sun Li, Quan Tu, and Deyi Xiong. Roleeval: A bilingual role evaluation benchmark for large language models, 2024. URL <https://arxiv.org/abs/2312.16132>.
- [34] Jisu Shin, Juhyun Oh, Eunsu Kim, Hoyun Song, and Alice Oh. Spotting out-of-character behavior: Atomic-level evaluation of persona fidelity in open-ended generation. In Wanxiang Che, Joyce Nabende, Ekaterina Shutova, and Mohammad Taher Pilehvar, editors, *Findings of the Association for Computational Linguistics: ACL 2025*, pages 26312–26332, Vienna, Austria, July 2025. Association for Computational Linguistics. ISBN 979-8-89176-256-5. doi: 10.18653/v1/2025.findings-acl.1349. URL <https://aclanthology.org/2025.findings-acl.1349/>.
- [35] Jen tse Huang, Wenxuan Wang, Eric John Li, Man Ho LAM, Shujie Ren, Youliang Yuan, Wenxiang Jiao, Zhaopeng Tu, and Michael Lyu. On the humanity of conversational AI: Evaluating the psychological portrayal of LLMs. In *The Twelfth International Conference on Learning Representations*, 2024. URL <https://openreview.net/forum?id=H3UayAQWoE>.
- [36] Quan Tu, Shilong Fan, Zihang Tian, Tianhao Shen, Shuo Shang, Xin Gao, and Rui Yan. CharacterEval: A Chinese benchmark for role-playing conversational agent evaluation. In Lun-Wei Ku, Andre Martins, and Vivek Srikumar, editors, *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 11836–11850, Bangkok, Thailand, August 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.acl-long.638. URL <https://aclanthology.org/2024.acl-long.638/>.
- [37] Lei Wang, Jianxun Lian, Yi Huang, Yanqi Dai, Haoxuan Li, Xu Chen, Xing Xie, and Ji-Rong Wen. CharacterBox: Evaluating the role-playing capabilities of LLMs in text-based virtual worlds. In Luis Chiruzzo, Alan Ritter, and Lu Wang, editors, *Proceedings of the 2025*

- Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 6372–6391, Albuquerque, New Mexico, April 2025. Association for Computational Linguistics. ISBN 979-8-89176-189-6. doi: 10.18653/v1/2025.naacl-long.323. URL <https://aclanthology.org/2025.naacl-long.323/>.
- [38] Noah Wang, Z.y. Peng, Haoran Que, Jiaheng Liu, Wangchunshu Zhou, Yuhan Wu, Hongcheng Guo, Ruitong Gan, Zehao Ni, Jian Yang, Man Zhang, Zhaoxiang Zhang, Wanli Ouyang, Ke Xu, Wenhao Huang, Jie Fu, and Junran Peng. RoleLLM: Benchmarking, eliciting, and enhancing role-playing abilities of large language models. In Lun-Wei Ku, Andre Martins, and Vivek Srikumar, editors, *Findings of the Association for Computational Linguistics: ACL 2024*, pages 14743–14777, Bangkok, Thailand, August 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.findings-acl.878. URL <https://aclanthology.org/2024.findings-acl.878/>.
- [39] Xintao Wang, Yunze Xiao, Jen-tse Huang, Siyu Yuan, Rui Xu, Haoran Guo, Quan Tu, Yaying Fei, Ziang Leng, Wei Wang, Jiangjie Chen, Cheng Li, and Yanghua Xiao. InCharacter: Evaluating personality fidelity in role-playing agents through psychological interviews. In Lun-Wei Ku, Andre Martins, and Vivek Srikumar, editors, *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1840–1873, Bangkok, Thailand, August 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.acl-long.102. URL <https://aclanthology.org/2024.acl-long.102/>.
- [40] Xintao Wang, Heng Wang, Yifei Zhang, Xinfeng Yuan, Rui Xu, Jen tse Huang, Siyu Yuan, Haoran Guo, Jiangjie Chen, Shuchang Zhou, Wei Wang, and Yanghua Xiao. CoSER: Coordinating LLM-based persona simulation of established roles. In *Forty-second International Conference on Machine Learning*, 2025. URL <https://openreview.net/forum?id=B0rR7YqKUt>.
- [41] Hongqiu Wu, Weiqi Wu, Tianyang Xu, Jiameng Zhang, and Hai Zhao. Towards enhanced immersion and agency for LLM-based interactive drama. In Wanxiang Che, Joyce Nabende, Ekaterina Shutova, and Mohammad Taher Pilehvar, editors, *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 11166–11182, Vienna, Austria, July 2025. Association for Computational Linguistics. ISBN 979-8-89176-251-0. doi: 10.18653/v1/2025.acl-long.546. URL <https://aclanthology.org/2025.acl-long.546/>.
- [42] Chen Xu, Piji Li, Wei Wang, Haoran Yang, Siyun Wang, and Chuangbai Xiao. Cosplay: Concept set guided personalized dialogue generation across both party personas. In *Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval, SIGIR ’22*, page 201–211. ACM, 2022. doi: 10.1145/3477495.3531957. URL <http://dx.doi.org/10.1145/3477495.3531957>.
- [43] Rui Xu, Xintao Wang, Jiangjie Chen, Siyu Yuan, Xinfeng Yuan, Jiaqing Liang, Zulong Chen, Xiaoqingdong, and Yanghua Xiao. Character is destiny: Can persona-assigned language mod-

els make personal choices? In Christos Christodoulopoulos, Tanmoy Chakraborty, Carolyn Rose, and Violet Peng, editors, *Findings of the Association for Computational Linguistics: EMNLP 2025*, pages 15038–15059, Suzhou, China, November 2025. Association for Computational Linguistics. ISBN 979-8-89176-335-7. doi: 10.18653/v1/2025.findings-emnlp.813. URL <https://aclanthology.org/2025.findings-emnlp.813/>.

- [44] An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, et al. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*, 2025.
- [45] Jinfeng Zhou, Zhuang Chen, Dazhen Wan, Bosi Wen, Yi Song, Jifan Yu, Yongkang Huang, Pei Ke, Guanqun Bi, Libiao Peng, JiaMing Yang, Xiyao Xiao, Sahand Sabour, Xiaohan Zhang, Wenjing Hou, Yijia Zhang, Yuxiao Dong, Hongning Wang, Jie Tang, and Minlie Huang. CharacterGLM: Customizing social characters with large language models. In Franck Dernoncourt, Daniel Preotiuc-Pietro, and Anastasia Shimorina, editors, *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing: Industry Track*, pages 1457–1476, Miami, Florida, US, November 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.emnlp-industry.107. URL <https://aclanthology.org/2024.emnlp-industry.107/>.

## A Character Arc Construction Details

**Why two streams.** A single end-to-end pass conflates two distinct failure modes: missing a psychologically impactful event, and misreading the character’s state given the same events. Running the event and state streams in parallel keeps the two failures separable in any downstream audit, and partial convergence between the streams serves as an internal reliability check. Each chapter is processed independently, with a recurrent summary that carries the prior chapter’s running state forward.

**Axis induction constraints.** Each character receives 3–5 intrapersonal axes plus one relational axis per target (a relational target is admitted only when the text exhibits non-trivial trackable change). The first intrapersonal axis is anchored to the agency–communion dimension [22, 23] as a sanity coordinate, and its absence in the induced set is a strong drift signal. Every induced axis must ground in established literary or psychological scholarship, following the construct-grounding criterion of *Values in the Wild* [14] (Figure 6).

**Reconciliation tag set.** Stream-unique candidates receive one of three tags, each tied to a distinct post-processing outcome (Figure 2, top middle). **valid\_missed** (*kept*) marks a candidate that is plausible on review but absent from the other stream’s set, and is added to the merged axis set. **borderline** (*flagged*) marks candidates with insufficient evidence to commit either way, which are retained but tagged for downstream filtering. **artifact** (*discarded*) marks candidates that are products of how the evidence was scoped (for example, a transient event-stream salience). Matched pairs receive an arc-direction label that extends McAdams’ redemptive / contaminating

Alexey Alexandrovich Karamzin — Anna Karamzina

Intergenerational

axis 2 / 8

annotator: 3/4

Annotation: 3/4

How to read this page — element guide

agency-communion

From formal self-reliance to emotional, then compassionate relational openness

confidence: high — source text: 100% — evidence: 100%

The color defines the direction and meaning of the axis — what psychological dimension it is measuring and the two ends of that spectrum. Use them to understand what change the axis claims to capture before reading the trajectory.

→ High (low) — beginning (end) of arc

High agency, low communion Karamzin begins as a self-contained bureaucratic husband and official who relies on formality, duty, and public norms to order feeling and maintain control.

→ High (low) — end of arc

He remains an obediently agency, but with a notable increase in communion he becomes capable of pity, forgiveness, tenderness, and spiritual magnanimity, though this openness says little and partly motivated by duty and religion.

↑ Trajectory

The trajectory shows the stages through which the character moves along this axis. Each phase includes a chapter range, a description of the character's position at that point, and key moments from the text as supporting evidence for that phase.

Initial formal agency

1-10-11

As begins as cool, controlled, and socially polished, expressing agency through etiquette, bureaucratic competence, and emotional distance rather than sympathy.

→ Key moments

Early emotional

12-13-14

Intuition, bias, and hesitation weaken his mastery; he still tries to manage events, but he internal life becomes more emotional and vulnerable.

→ Key moments

Formal, practical agency

15-16-17

He correctly deploys his administrative abilities: legal consultation, custody concerns, and formal representing moral agency to his superior and subordinates rather than sympathy.

→ Key moments

Most selfless and spiritual magnanimity

18-19-20

His agency becomes less service and more responsive to his experience pity, forgiveness, tenderness, and religious inspiration toward others. He helps, and even forgives.

→ Key moments

Less self-reliance

21-22-23

He ends in diminished self-reliance and is vulnerable to public and paternal rules, but inwardly motivated by tenderness and emotional communion rather than duty and sympathy.

→ Key moments

↓ Evidence summary

A summary of the specific evidence in the novel that support this axis. Use this to verify whether the axis is grounded in actual events from the text — does the evidence match your reading of the novel?

Karamzin starts as a highly agency, low communion figure and is repeatedly broken open by betrayal, humiliation, grief, and religion. These experiences increase his capacity for communion — especially pity, forgiveness, and tenderness — without fully displacing his need for order and control.

→ Library Evidence — external scholarly citations

👤 Your Rating

Based on the poles, trajectory, and evidence summary above, please answer the following questions. You can skip this axis — your rating is optional.

Q1: Does this axis accurately reflect a real dimension of change for this character?

☒ Yes ☐ No

Q2: Does this axis run through the entire novel?

☒ Yes — present and develops throughout the entire novel ☐ No — confined to a specific part of the novel and does not carry through

Q3: How important is this axis to understanding the character?

☐ 1 — Negligible ☐ 2 — Minor ☐ 3 — Moderate ☐ 4 — Important ☐ 5 — Central to character

Q4: How confident are you in your rating above?

☐ 1 — Very uncertain ☐ 2 — Somewhat uncertain ☐ 3 — Moderately confident ☐ 4 — Confident ☐ 5 — Very confident

Q5: Add comment (optional)

← Previous

Axis 6 Next

→ Skip (optional)

FIGURE 5 Human Annotation Page for axis rating.

dichotomy to non-linear trajectories such as oscillating or U-shaped arcs.

**LLM-critic ensemble.**The Stage iii critics adopt three literary perspectives: a structuralist / narratologist critic that reads the axis against the novel’s plot architecture, a depth-psychological critic that reads it against psychoanalytic and developmental frameworks, and a historical / cultural critic that reads it against the novel’s social-historical context. Each critic attaches scholarly citations as candidate evidence, not as ground truth. The citations are passed to human annotators alongside the axis for their own use.

**Human-annotator ratings.**Three annotators independently rate each axis on three substantive dimensions (Figure 2, top right): *validity* (does the axis describe a real psychological or relational change in the source text?), *novel span* (the proportion of the novel over which the change unfolds), and *importance* (the centrality of the change to the character’s portrayal). Each rating is accompanied by a *confidence* self-rating that records the annotator’s certainty in their three substantive judgments. Confidence is a meta-rating rather than a property of the axis. Only validity gates evaluation-set entry, with a 2-of-3 majority rule. The remaining ratings are stored as per-axis metadata so analyses can stratify, for example reporting fidelity separately on high- vs. low-importance axes. The axis-validity annotation did not constitute human subjects research under our institution’s IRB definitions. The task asked annotators to judge whether each extracted character-arc axis is faithful to the source novel. No personally identifiable information, sensitive data, or behavioral data about the annotators themselves was collected. Annotators were volunteers who self-reported having read the target novel and were compensated at a fair rate; all participation was on the basis of informed consent.(see Figure 5)





**FIGURE 6** Induced axes are clustered and grounded against established literary or psychological scholarship, following the construct-grounding criterion of *Values in the Wild* [14].

## B Probe Generation Details

**Out-of-World era menu.** The era menu has seven slots: *modern urban*, *industrial early 20th century*, *mid-century*, *pre-industrial agrarian*, *speculative near-future*, *pre-modern imperial*, and *diasporic contemporary*. Each slot is broad enough to generalize across source novels yet specific enough to commit the designer to one setting (suppressing a default contemporary-urban collapse seen in early pilots). An (axis, phase) pair is assigned an era by a cyclic walk through the menu starting from a SHA-1-derived offset of the axis id, which guarantees that all  $N$  phases of one arc receive distinct eras when  $N$  does not exceed the menu size.

**Per-(scene phase, probe type) structure.** For an arc of  $N$  phases, the designer is called  $3N$  times: once per (scene phase, probe type) cell. The scene phase is the trajectory phase whose lived experience grounds the scenario, and the other  $N - 1$  phase responses in the same probe are counterfactual projections of those phases’ policies onto that scenario. This cell is also the unit at which Stage iii validators act: per-response validators score each of the  $N$  responses in a cell, and per-probe validators score the cell as a whole.

**Q-PhaseFit verdict mapping.** Q-PhaseFit returns the trajectory phase index  $\hat{k}$  that the response is most diagnostic of, plus a confidence label. The wrapper compares  $\hat{k}$  to the target phase  $k^\star$  and emits one of three verdicts: **pass** ( $\hat{k} = k^\star$ ), **adjacent** ( $|\hat{k} - k^\star| = 1$ ), or **off\_phase** ( $|\hat{k} - k^\star| \geq 2$ ). Only **off\_phase** triggers the per-response regeneration path. **adjacent** is recorded but not corrected, since adjacent overlap is theoretically expected when adjacent phases sit close in the state distribution [9].

**Drop and retry policy.** Post-processing is asymmetric. A per-response failure (Q-Voice **fail**, or Q-PhaseFit verdict **off\_phase**) triggers one regeneration of that response. If the regenerated response also fails, it is marked **unavailable** and the rest of the probe is kept. A per-probe failure drops the whole probe: Q-Anchor failure for *In-Scenario*, Q-World failure for *In-World* and *Out-of-World*. Q-Discrim is annotation-only and never drops a probe, so adjacent-pair similarity surfaces as a Stage iii covariate rather than as a filter.

## C Datapoint information

The ARCANÉ corpus serialises into two record types: a *Character Arc* record per (character, axis), and a *probe* record per (axis, anchor phase, probe category). Probe records carry  $N$  *phase responses* as nested objects, one per phase of the parent arc. Both record types live in JSON and share a common key schema across the training and evaluation subsets of Table 3; the evaluation subset arcs additionally carry the Stage iii critic verdicts and the three-annotator validity votes (§3.1).

**Character Arc record.** Each arc fixes one axis for one character and is the unit indexed by `axis_id`. The header fields are `character`, `axis_type` (`intrapersonal` or `relational`), `dimension_label` (the McAdams construct family, e.g. *Agency-Communion*), `axis_name`, `pole_start` and `pole_end` (one-sentence descriptions of the two poles), and `arc_direction` (*redemptive*, *contaminating*, *ambivalent*, etc.; see Appendix A). `source` marks whether the candidate was emitted by the event stream, the state stream, or both (§3.1). The `trajectory` field is an ordered list of  $N$  phase objects (`phase`, `chapter_range`, `position_description`, `key_moments`). For relational arcs, `target_character` names the second party. Evaluation subset arcs additionally carry `literary_validation`, a triplet of critic verdicts (structuralist, depth-psychological, historical/cultural) each paired with reasoning and candidate citations, and `annotation`, a per-annotator vote record over *validity*, *novel span*, and *importance*.

**Probe record.** Each probe pairs one (scenario, question) with  $N$  phase-keyed reference responses. The header fields are `probe_id`, `probe_type` ( $\in \{In-Scenario, In-World, Out-of-World\}$ ), `anchor_phase_idx` (the trajectory phase whose lived experience grounds the scenario), `anchor_query_chapter` (the chapter at which the anchor response is set, which also caps the knowledge cutoff), `scenario`, and `question`. *In-Scenario* probes carry an `anchor_source` pointer back to the verbatim passage in the source novel; *Out-of-World* probes carry `era_label` together with the `abstract_axis` and `abstract_phases` reformulations that let the axis transpire into a non-source era (Appendix B). The `phase_responses` list contains  $N$  objects, one

```

probe_id: anna_karenina_final_intra_01_oow_a0
probe_type: out_of_world  anchor_phase_idx: 0  anchor_query_chapter: 20
era_label: pre_industrial_agrarian
abstract_axis: "how self-assertion shifts under relational strain from prosocial attunement toward coercive control
and ultimately self-negating withdrawal"
scenario: "At the harvest supper, Anna sees two households quarrel over a promised strip of meadow, and the man
whose favor matters most to her stands waiting to see whether she will speak for peace or for herself. The room falls
quiet around her."
question: "How does Anna answer the silence in the room?"
phase_responses[0] (Socially fused mediator, ch. 20)
  gt_action: "She steps between the benches and proposes that the meadow be shared until the elders measure it
after harvest."
  gt_thought: "She feels the strain in every face and moves to ease it; if she can give each side a little dignity, the
whole gathering may hold together."
phase_responses[1] (Divided self and concealed attachment, ch. 36)
  gt_action: "She lowers her eyes and offers a mild compromise, careful not to meet the favored man's gaze for
long."
  gt_thought: "She wants peace, yet her own hidden leaning makes every word feel stained; if she speaks too warmly,
others may hear more in it than the dispute itself."
(phases 2-4 omitted for space)

```

**FIGURE 7** One *Out-of-World* probe from Anna Karenina’s *Agency–Communion* arc, truncated to the first two of  $N=5$  phase responses. The same (scenario, question) is asked at every phase, with the references shifting along the axis.

| Split                  | #Novels | #Chars | #Arcs | #Probes | #SFT   |
|------------------------|---------|--------|-------|---------|--------|
| Train                  | 10      | 48     | 308   | 2,545   | 45,690 |
| Test (validated)       | 5       | 25     | 205   | 1,754   | –      |
| Test (unval., low-pop) | 2       | 7      | 31    | 302     | –      |
| Total                  | 17      | 80     | 544   | 4,601   | 45,690 |

**TABLE 3** Dataset statistics. *Train* uses ungrounded auto arcs. *Test (validated)* uses human-validated arcs (§3.1). *Test (unval., low-pop)* uses auto arcs on two low-popularity unvalidated novels as a memorization control [12] (§4.4). SFT data is generated only for Train.

per phase, each with `phase_idx`, `phase_label`, `query_chapter`, `gt_action` (the overt act), `gt_speech` (the in-character utterance, when one is licensed), `gt_thought` (one to two sentences of cognitive construal), and `gt_typicality` ( $\in \{\textit{typical}, \textit{plausible\_tail}\}$ ). Stage iii validator outputs are carried alongside the responses: per-response verdicts from Q-Voice and Q-PhaseFit sit inside each phase response under `quality`, with `retry_count` and `unavailable` recording the drop-and-retry trace (Appendix B); per-probe Q-Anchor / Q-World and Q-Discrim verdicts sit at the probe level under `probe_check`, and Q-Discrim-flagged adjacent pairs are mirrored on the parent arc under `weak_pairs`.

**Example.** Figure 7 shows a single *Out-of-World* probe drawn from Anna Karenina’s *Agency–Communion* arc, abbreviated to the first two of  $N=5$  phase responses. The shared scenario is asked at every phase, and the phase-keyed reference responses encode how the same character would handle that scene at successive points on the axis.

**Artifact availability and licenses.** The source texts are drawn from Project Gutenberg and are in the U.S. public domain. The released ARCANE artifacts (character arcs, probes, validator outputs, and accompanying code) will be distributed under CC-BY-4.0 for data and MIT for code. The open-weight models used for inference and fine-tuning (e.g., Qwen3, DeepSeek-V4 family) are used under their respective publishers’ licenses, and the proprietary models accessed via API (§4.1, Appendix F) are used under the providers’ terms of service. The released ARCANE artifacts are intended for non-commercial research on context-grounded role-playing language agents, consistent with the original access conditions of the source materials.

## D Experimental Details

### D.1 Datasets

This section provides details for the datasets. Both SFT and DPO datasets are derived from the ARCANE corpus and share the same underlying hierarchy, differing only in how they are formulated.

**SFT dataset.** The 45,690 SFT rows reported in Table 3 are generated on the 10 training novels and cover only the two trajectory-aware context modes, *Arc* and *Summary*. For each (probe, phase, mode) triplet, we sample three completions from each of two models, `gpt-5.4-mini` (OpenAI) and `claude-sonnet-4-6` (Anthropic), with  $T = 0.9$  and a 800-token cap. Pooling two models reduces single-teacher idiosyncrasy in the SFT distribution.

**DPO dataset.** Preference pairs are constructed automatically, which spans 55 characters across 12 novels, organized along 113 character-development axes and 2,838 unique probes in ARCANE corpus. For each probe, we designate the anchor phase response as chosen and a response from an adjacent phase as rejected, with both sides sharing the same scenario and question. This ensures that the only contrast between chosen and rejected is the character’s behavioral disposition at different developmental stages, rather than differences in scenario or context. The resulting training set contains 14,671 preference pairs across 2,516 probes.

### D.2 Two-Stage Pipeline

This section provides details for the two-stage training pipeline described in §3.4, applied to both ARCANE-8B and ARCANE-32B. The two stages are designed to be complementary. SFT instills the form and texture of ARCANE corpus, and the DPO stage sharpens the model’s sensitivity to developmental trajectory by contrasting anchor phase behavior against plausible adjacent phase alternatives.

**Stage 1: Supervised fine-tuning.** In the SFT stage, the model learns to produce responses that conform to the format and behavioral conventions of ARCANE, which differ substantially from standard instruction-following. Generations from two models (`gpt-5.4-mini` and `claude-sonnet-4-6`) serve as imitation targets.

**Stage 2: Direct preference optimization.** In the DPO stage, the model learns to distinguish ground-truth character behavior from plausible but temporally displaced alternatives, namely, actions or thoughts that fit the character but belong to a different narrative phase.

### D.3 Training Configuration

This section provides the training configuration. Table 4 summarizes the training configurations for both ARCANE-8B and ARCANE-32B models across the two-stage pipeline.

|                      | ArcANE-8B                      | ArcANE-32B |
|----------------------|--------------------------------|------------|
| Base model           | Qwen3-8B                       | Qwen3-32B  |
| Fine-tuning          | Full                           | LoRA       |
| <i>SFT</i>           |                                |            |
| Epochs               | 1                              | 1          |
| Learning rate        | 1e-5                           | 1e-4       |
| Effective batch size | 64                             | 32         |
| Max length           | 8192                           | 8192       |
| Training time        | 2h 38m                         | 9h 49m     |
| <i>DPO</i>           |                                |            |
| Epochs               | 1                              | 1          |
| Learning rate        | 5e-6                           | 1e-5       |
| Batch size           | 64                             | 64         |
| Max length           | 8192                           | 8192       |
| Training time        | 3h 44m                         | 14h 3m     |
| <i>LoRA</i>          |                                |            |
| Rank ( $r$ )         | —                              | 64         |
| Alpha ( $\alpha$ )   | —                              | 128        |
| GPU                  | 1 $\times$ NVIDIA B200 (192GB) |            |

**TABLE 4** Training configuration for ARCANE-8B and ARCANE-32B, following the official Qwen3 sampling recommendations for non-thinking mode.

### D.4 Inference Details

This section describes the inference setup used for all models evaluated in §4.

**Inference setup and cost.** Open-weight models (Qwen3-8B, Qwen3-32B, ARCANE-8B, and ARCANE-32B) are served locally via **vLLM** on a single NVIDIA B200 (192GB) GPU, following the official Qwen3 sampling recommendations for non-thinking mode. Generation is capped at

8,192 tokens, set at the 99th percentile of observed response lengths, to guard against runaway generations. Covering all novels and modes, inference takes up to 6 hours per model.

Open-weight large models (DeepSeek-V4-Flash and DeepSeek-V4-Pro) are called through `OpenRouter` with the provider pinned to a fixed pair (`DeepSeek, Alibaba, allow_fallbacks=false`) so that role-playing calls cannot silently re-route to a different quantization.

Role-play generation passes no explicit decoding parameters; each backend’s default sampling applies (effectively temperature 1.0 with no top- $p$  or repetition-penalty restriction across vLLM, OpenAI, and OpenRouter). One sample is drawn per (probe, mode, model). Two subcalls override this for determinism: the TimeCHARA two-stage hint extractor and the LLM judge (§4.2) both run at  $T = 0$ .

## E Context-Mode Templates

The six context modes evaluated in §4.1 share the same (scenario, question, query chapter) prompt body and differ only in the system-side context block.

**Vanilla.** The system message states the character’s identity and the query chapter, and no further context is supplied. Vanilla is the reference for how much the LLM can do from its own world knowledge.

**Summary.** Vanilla plus per-chapter summaries for the most recent `SUMMARY_MAX_CHAPTERS = 5` chapters up to and including the query chapter. The summaries are produced once per novel by the `summary` pipeline of `experiments/`.

**RAG.** Vanilla plus the top- $k$  source-text chunks retrieved with the (scenario, question) string as the embedding query. Chunks are 1.5k-character windows with 300-character overlap, embedded with `text-embedding-3-small`, and retrieval is masked to chunks up to the query chapter, with  $k = 6$ .

**LifeChoice.** A port of CHARMAP [43]. The *description* is the chapter-summary stack (the same source as Summary), and the *memory* is a top- $k$  retrieval over the RAG index using the description as the embedding query (CHARMAP’s contribution over plain RAG).

**TimeCHARA.** A port of the narrative-experts pipeline of Ahn et al. [2]: a first call predicts the query chapter and the character’s presence in it, and the second call uses those predictions as inline hints to the role-play model. No system-side context block is added.

**Arc (ours).** Vanilla plus the curated character arc up to the query chapter. The payload is the final-stage axis JSON with phases beyond the query chapter hidden. `literary_validation` and `evidence_summary` are always stripped (the former is the human-rater bibliography, the latter summarises the whole arc and would leak future state). When at least one later phase is hidden, `pole_end` and `arc_direction` are also removed so the truncated payload cannot reveal where the trajectory is heading.



## F Judge Setup and Rubrics

**Judge model and serving.** The default judge is `deepseek/deepseek-v4-flash` served through OpenRouter, with the provider pinned to a fixed pair (`DeepSeek, Alibaba`) and `allow_fallbacks=false`, so judge calls cannot silently re-route to a different provider or quantization. The reasoning pass is disabled (`EXP_JUDGE_DISABLE_REASONING=true`) because each verdict is a small JSON object and the thinking pass otherwise dominates the bill ( $\sim 57\times$  the static prompt). Each run uses a 200-thread judge pool, with per-score scale 1–100. The judge sees both `ref_action` and `ref_thought` by default. A no-thought variant (`INCLUDE_THOUGHT=false`) instructs the judge to infer the reference reasoning from `ref_action` and `ref_speech` alone, used as an ablation of the rubric’s dependence on `ref_thought`.

**Per-response rubric (APF / RPF / RAE).** Each (probe, phase) response is scored on three sub-dimensions. *APF* judges whether the response’s action is mechanism-equivalent to `ref_action` at the target phase, with three salvage levels (A strict, B and C giving partial credit when A fails). *RPF* parses `ref_thought` and the response’s reasoning into four mechanism slots (appraisal, goal, strategy, trigger) and judges each slot, then aggregates. *RAE* fixes `ref_thought` as the operative reasoning and asks whether the response’s action is one this reasoning would license. `gt_entailment` is the dominant sub-check, so a response with internally consistent but `ref_thought`-incompatible reasoning still scores low.

**Trajectory rubric (PTF).** The trajectory judge sees the scenario, the question, and  $N$  phase blocks in chronological order. Each block carries the reference fields and the model’s response at that phase. PTF aggregates three sub-scores (alignment, direction, shape), each on the 1–100 scale, designed so a model that scores well on every phase in isolation can still earn a low PTF by collapsing adjacent phases, traversing the axis in the wrong direction, or compressing the shape of the change. `unavailable` phase responses are dropped from their blocks, but the original phase indices are preserved so the judge can see the gap.

The judge-side no-thought ablation swaps the two system prompts for variants that drop the `ref_thought` mention and instruct the judge to infer the reference reasoning from `ref_action` and `ref_speech` alone. The sub-check definitions are unchanged.

## G Reward-Hacking Falsification

The reward-hacking probe in §5.3 runs two structured pair-trace procedures on the same stratified 150-probe sample: a four-proposition coding test and a POV-control regeneration. The four propositions were not chosen *a priori*. An author close-reading of six paired trajectories surfaced them first. We describe the close-reading, then the proposition test, then the POV control, then the joint result.

**Coder configuration.** The proposition test uses `deepseek/deepseek-v4-pro` on OpenRouter (provider pinned to `DeepSeek`, `allow_fallbacks=false`, temperature 0, `json_object` response format, 16–32-thread `ThreadPoolExecutor`). We use a pro-tier coder rather than the *flash*-tier

judge that scores APF/RPF/RAE/PTF (Appendix F) because each call reads two full per-phase response sets ( $\sim 2\text{--}5$  phases per side,  $\sim 5\text{--}15\text{K}$  input tokens) and emits a structured verdict.

**Sampling.** The pair-trace pool is the validated slice (1,750 probes, 4 dropped for missing judge scores). For each probe we compute per-probe Overall under Arc as the mean over  $\{\text{APF, RPF, RAE, PTF}\}$ , with APF/RPF/RAE averaged across the probe’s phase responses and PTF the mean of the three trajectory sub-scores. Per-probe deltas are tagged *positive* ( $\Delta > 1$ ), *tie* ( $|\Delta| \leq 1$ ), or *negative* ( $\Delta < -1$ ) on the 0–100 scale, and ties are dropped from coding. We stratify by (probe-type, sign) into six cells and draw 25 per cell (150 probes total, Qwen3-32B vs. ARCANE-DPO) for both the proposition test and the POV control.

**Close-reading six paired trajectories.** We close-read six probes spanning all three categories and both delta signs to surface candidate patterns: Dulcinea silent under courtly mockery (In-Scenario, relational), Don Quixote’s volunteer-patrol arc (Out-of-World, intrapersonal), Sansón Carrasco listening to Sancho (Out-of-World, relational), Darya Oblonskaya through a domestic-crisis cascade (In-World, intrapersonal), Snape after the Lily slur (In-Scenario, relational), and Karenin handling a clerk’s wife in public (In-World, intrapersonal). Four patterns recurred across all six.

- **P1, Phase distinctness.** Per-phase content is substantively different across phases versus paraphrase. ARCANE writes different content per phase. Qwen3-32B often paraphrases: on Carrasco–Sancho it opens *every* phase with “*Ah, Sancho Panza, you speak with the boldness of a man who has long since learned...*”. ARCANE moves him from amused engagement (“*Ha, Sancho, you speak as if you had a warrant in your sleeve!*”) to practical incorporation (“*I took it as a thing to be weighed, not mocked*”).
- **P2, Canonical specificity.** The response invokes named secondary characters, named places, or source-specific scenes, versus generic trait language. ARCANE’s Snape phase 1 routes through Dumbledore and the memory transfer. Qwen3-32B stays at a general lament (“*I would have chosen you, Lily*”).
- **P3, Negative-space fidelity.** When canon calls for silence, proxy, or absence, the response preserves it. Both Qwen3-32B and ARCANE pass Dulcinea: Qwen3-32B by third-person narration, ARCANE by routing through canon’s Merlin pageant. POV-Qwen (the register-controlled Qwen run, below) fails this case, inventing speech the canonical scene refuses.
- **P4, Phase register switch.** Voice and register shift to match each phase’s psychological description. ARCANE’s Darya opens phase 0 (“Emotional collapse”) with “*I feel my heart give a violent jump*” and closes phase 3 (“Late steady endurance”) with “*first care, then order, then the business of the day*”. Qwen3-32B keeps one composed-narrator register throughout.

**Proposition test.** For each of 150 probes  $\times$  3 model conditions (450 calls) the coder reads axis, scenario, question, reference fields (`ref_action`, `ref_thought`), and one model’s per-phase response set, and returns yes/no on each of P1–P4 with a one-sentence reason. P3 is tagged **na** when canon does not call for silence or proxy. The three conditions are: Qwen3-32B with the original system prompt, Qwen3-32B with one POV instruction appended (POV-Qwen, defined

next), and ARCANE-32B-DPO.

**POV-control regeneration.** For each (probe, phase) in the 150-probe sample we load the byte-identical `prompt_system` and `prompt_user` from the main paper’s arc run and append one paragraph:

*Respond IN CHARACTER, in the first person, in the present tense. Speak AS this character, not ABOUT them. Do NOT narrate from the outside; do not write ‘X would respond’ or ‘X’s manner is one of’; speak as the character. Use period-appropriate diction natural to the source novel. Do not mention ‘axes’, ‘phases’, or any meta-analytic framing.*

Calls go to `qwen/qwen3-32b` on OpenRouter pinned to the Alibaba provider with thinking disabled, the same backend used in the main runs. The resulting 509/510 phase responses are scored with the unchanged Stage-iii APF/RPF/RAE judge (`deepseek/deepseek-v4-flash`), prompts and provider settings identical to Appendix F.

**Results.** Table 5 reports proposition yes-rates by category. The Overall row collapses the three categories. Per-category rows are diagnostic. Bold marks the highest *when the proposition is category-appropriate*. For P2, “yes” on In-World and Out-of-World by POV-Qwen is largely context-violating canon-drop (e.g. Voldemort name-dropped into a city transit hub), not honest specificity, so we do not bold those cells. P3 is applicable on only 2 probes in this stratified sample (the silence-or-proxy structure is rare on the validated slice), so we report it qualitatively above.

**Rubric scores and a representative case.** On the same 150 probes, the Stage-iii APF/RPF/RAE rubric judge scores Qwen3-32B 53.8, POV-Qwen 50.0, and ARCANE-32B-DPO 56.7 (per-probe Overall, 0–100). The POV instruction lowers the rubric score by penalising context-violating canon-drop. The Dulcinea phase-1 probe (“*Theatrical prop and staged manipulation*”) is representative. Qwen3-32B writes “*Dulcinea...does not speak, does not interject*” in third person. ARCANE routes through canon’s Merlin pageant (“*Dulcinea answered in the person of another...a squire in a mantle of green*”). POV-Qwen, forced into first person, invents a speech canon refuses (“*‘Beauty, that fleeting muse of mortal eyes...’*”). Rubric scores: 80, 40, 22 respectively. The same pattern, POV-Qwen losing on fabrication while ARCANE-32B-DPO stays anchored, recurs across the high-drop tail.

**Note on a discarded discovery pass.** Before the close-reading above, we ran a free-form LLM coding pass on a 72-probe sample (8 per (probe-type, sign) cell from Qwen3-32B vs. ARCANE-DPO, 4 per cell from ARCANE-SFT vs. ARCANE-DPO). The coder returned per-probe `summary`, `trajectory_shape`, 2–4 `concrete_tells`, a `substance_or_style` tag, and a 4–10 word `category_label`. A second LLM call clustered the observations into themes. The themes confounded POV style with substantive arc movement, so we do not use them in the main paper. The author close-reading above replaced the failed clustering.

| Proposition              | Category     | Qwen3-32B | POV-Qwen | ARCANE-DPO  |
|--------------------------|--------------|-----------|----------|-------------|
| P1 Phase distinctness    | In-Scenario  | 4.5       | 61.4     | 53.3        |
|                          | In-World     | 22.0      | 70.0     | <b>74.0</b> |
|                          | Out-of-World | 4.1       | 55.1     | <b>67.3</b> |
|                          | Overall      | 10.5      | 62.2     | <b>65.3</b> |
| P2 Canonical specificity | In-Scenario  | 20.0      | 68.0     | <b>34.0</b> |
|                          | In-World     | 8.0       | 60.0     | 2.0         |
|                          | Out-of-World | 4.0       | 54.0     | 2.0         |
|                          | Overall      | 10.7      | 60.7     | 12.7        |
| P4 Phase register switch | In-Scenario  | 6.8       | 61.4     | 53.3        |
|                          | In-World     | 22.0      | 72.0     | <b>76.0</b> |
|                          | Out-of-World | 4.1       | 57.1     | <b>65.3</b> |
|                          | Overall      | 11.2      | 63.6     | <b>65.3</b> |

**TABLE 5** Proposition yes-rates (% , excluding **na**) on  $n=150$  probes per cell, judge **deepseek-v4-pro**. Per-category cells  $n \approx 50$ .

## H SFT vs DPO Close-Coding

What does DPO add over SFT (§5.3)? The ARCANE-SFT vs. ARCANE-DPO pair-trace covers 1,750 probes (mean  $\Delta=+1.89$ ,  $N_+=940$ ,  $N_-=702$ ). We close-code 180 of them (30 per (probe-type, sign) cell) using the same free-form LLM coder as the discarded discovery pass in Appendix G. On the SFT→DPO pair the resulting themes are coherent rather than confounded, dominated by *phase-driven arc vs. static register*: ARCANE-DPO sharpens what ARCANE-SFT already started.

**Two illustrative cases.** On Don Quixote’s interdependence axis, ARCANE-SFT keeps a single knightly tag (“*Stand aside, good people, I go in service of order*”) through every phase. ARCANE-DPO drops the knightly voice in the repentant phase and turns toward family (“*You need not fear for me...come home, and let us settle what ought to be settled in peace*”). On Voldemort, ARCANE-SFT keeps him stoic across phases (“*I do not falter*”). ARCANE-DPO finds the fracture (“*‘No!’ I snarled, and the word itself seemed to recoil from me*”).

**One regression.** On Hagrid the ARCANE-DPO compression hurts. Hagrid’s idiom is communal, dramatically expansive, and multi-speaker, and it sits more naturally in Qwen3-32B’s third-person scene narration than in ARCANE-DPO’s first-person interior monologue. This is the only character in the 180-probe sample where the first-person register ARCANE installs is a net loss.

| Family      | Size  | Mode       | In-Scenario |             |             |             | In-world    |             |             |             | Out-of-world |             |             |             | Overall     |
|-------------|-------|------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|--------------|-------------|-------------|-------------|-------------|
|             |       |            | APF         | RPF         | RAE         | PTF         | APF         | RPF         | RAE         | PTF         | APF          | RPF         | RAE         | PTF         | Overall     |
| DeepSeek-V4 | Flash | Vanilla    | 56.9        | 57.7        | 47.9        | 51.3        | 55.6        | 56.6        | 47.9        | 47.1        | 52.4         | 53.4        | 43.7        | 47.8        | 51.5        |
|             |       | Summary    | 60.6        | 60.2        | 52.8        | 56.9        | 54.5        | 55.5        | 45.9        | 48.6        | 51.9         | 52.9        | 42.4        | 46.4        | 52.4        |
|             |       | RAG        | 62.7        | 61.9        | <b>55.6</b> | 59.7        | 54.0        | 55.4        | 44.9        | 47.5        | 53.2         | 53.8        | 43.5        | 48.7        | 53.4        |
|             |       | LifeChoice | 61.0        | 60.5        | 53.3        | 52.8        | 55.7        | 57.0        | 46.9        | 48.4        | 51.9         | 53.2        | 42.7        | 46.8        | 52.5        |
|             |       | TimeChara  | 56.6        | 57.3        | 48.6        | 51.8        | 53.1        | 54.2        | 45.1        | 43.1        | 53.8         | 54.8        | 45.2        | 45.8        | 50.8        |
|             |       | Arc        | <b>63.4</b> | <b>63.0</b> | 55.3        | <b>60.9</b> | <b>62.8</b> | <b>62.4</b> | <b>54.9</b> | <b>57.0</b> | <b>62.4</b>  | <b>62.3</b> | <b>54.9</b> | <b>53.6</b> | <b>59.4</b> |
|             | Pro   | Vanilla    | 54.3        | 55.2        | 45.9        | 51.1        | 53.0        | 54.2        | 45.6        | 45.3        | 52.5         | 53.5        | 44.0        | 43.0        | 49.8        |
|             |       | Summary    | 60.0        | 60.2        | 52.3        | 55.7        | 55.3        | 56.4        | 47.6        | 50.5        | 55.9         | 56.3        | 47.9        | 49.3        | 54.0        |
|             |       | RAG        | 64.1        | 63.1        | 57.6        | 58.4        | 55.8        | 56.6        | 48.6        | 46.3        | 52.9         | 53.7        | 44.3        | 47.0        | 54.0        |
|             |       | LifeChoice | 64.4        | <b>63.8</b> | <b>57.8</b> | 58.7        | 57.6        | 57.9        | 49.7        | 54.6        | 56.0         | 56.7        | 48.2        | 48.5        | 56.2        |
|             |       | TimeChara  | 54.9        | 55.6        | 46.8        | 50.4        | 51.4        | 52.4        | 43.8        | 43.0        | 50.5         | 51.6        | 42.1        | 44.3        | 48.9        |
|             |       | Arc        | <b>64.7</b> | 63.6        | 57.1        | <b>60.5</b> | <b>64.3</b> | <b>64.1</b> | <b>58.8</b> | <b>57.7</b> | <b>63.9</b>  | <b>64.0</b> | <b>56.9</b> | <b>56.2</b> | <b>61.0</b> |
| Qwen3       | 8B    | Vanilla    | 35.5        | 36.7        | 26.1        | 27.8        | 35.6        | 36.8        | 26.7        | 25.9        | 36.0         | 37.2        | 26.6        | 28.6        | 31.6        |
|             |       | Summary    | 43.3        | 42.3        | 33.0        | 33.5        | 39.4        | 39.7        | 29.8        | 32.9        | 36.5         | 36.8        | 26.4        | 28.1        | 35.1        |
|             |       | RAG        | <b>49.2</b> | <b>48.0</b> | <b>39.8</b> | 37.7        | 38.5        | 38.7        | 28.4        | 30.3        | 35.6         | 36.1        | 25.8        | 27.7        | 36.3        |
|             |       | LifeChoice | 47.2        | 45.6        | 37.0        | <b>42.0</b> | 39.5        | 39.4        | 29.1        | 30.2        | 37.9         | 38.0        | 27.8        | 29.0        | 36.9        |
|             |       | TimeChara  | 38.7        | 40.3        | 29.5        | 31.0        | 38.2        | 39.3        | 29.1        | 29.0        | 37.4         | 38.4        | 27.8        | 27.7        | 33.9        |
|             |       | Arc        | 46.7        | 46.6        | 36.1        | 37.5        | <b>44.6</b> | <b>45.2</b> | <b>33.9</b> | <b>37.6</b> | <b>46.2</b>  | <b>46.3</b> | <b>35.9</b> | <b>35.3</b> | <b>41.0</b> |
|             | 32B   | Vanilla    | 40.1        | 40.4        | 30.0        | 33.2        | 40.8        | 41.8        | 31.8        | 30.2        | 41.4         | 42.8        | 31.9        | 31.6        | 36.3        |
|             |       | Summary    | 50.5        | 50.0        | 40.3        | 42.7        | 46.4        | 46.9        | 36.8        | 36.5        | 44.3         | 45.0        | 34.3        | 36.0        | 42.5        |
|             |       | RAG        | <b>55.2</b> | 53.2        | 45.4        | 41.3        | 46.2        | 47.0        | 36.4        | 36.7        | 42.9         | 42.9        | 32.9        | 33.0        | 42.8        |
|             |       | LifeChoice | 54.8        | <b>54.1</b> | <b>46.0</b> | 45.9        | 46.8        | 47.8        | 37.2        | 35.1        | 43.8         | 44.4        | 34.2        | 37.3        | 44.0        |
|             |       | TimeChara  | 42.6        | 42.9        | 32.7        | 34.8        | 41.0        | 42.0        | 31.7        | 29.6        | 42.3         | 43.1        | 32.4        | 32.0        | 37.3        |
|             |       | Arc        | 55.1        | 53.5        | 44.0        | <b>50.7</b> | <b>55.1</b> | <b>54.7</b> | <b>45.6</b> | <b>44.2</b> | <b>53.1</b>  | <b>52.6</b> | <b>43.3</b> | <b>42.3</b> | <b>49.5</b> |
| ARCANE      | 8B    | Vanilla    | 44.2        | 43.6        | 34.9        | 38.3        | 46.1        | 46.2        | 37.2        | 34.6        | 43.1         | 43.0        | 34.0        | 35.0        | 40.0        |
|             |       | Summary    | 52.8        | 51.4        | 44.5        | 46.1        | 45.2        | 45.3        | 36.2        | 40.3        | 44.5         | 44.7        | 35.7        | 36.0        | 43.6        |
|             |       | RAG        | <b>59.1</b> | <b>57.6</b> | <b>51.9</b> | 51.2        | 48.5        | 48.6        | 39.7        | 38.5        | 45.4         | 45.5        | 36.8        | 35.1        | 46.5        |
|             |       | LifeChoice | 53.7        | 52.6        | 45.0        | 49.0        | 44.9        | 45.2        | 35.8        | 38.3        | 44.7         | 45.0        | 36.4        | 37.3        | 44.0        |
|             |       | TimeChara  | 44.4        | 45.1        | 36.0        | 38.6        | 45.8        | 46.3        | 37.2        | 34.8        | 43.7         | 43.9        | 35.1        | 35.7        | 40.6        |
|             |       | Arc        | 58.5        | 57.2        | 50.4        | <b>51.6</b> | <b>60.8</b> | <b>58.8</b> | <b>52.1</b> | <b>55.4</b> | <b>62.0</b>  | <b>59.7</b> | <b>54.4</b> | <b>51.6</b> | <b>56.0</b> |
|             | 32B   | Vanilla    | 48.9        | 49.3        | 39.9        | 41.6        | 55.3        | 55.5        | 47.0        | 41.9        | 55.9         | 55.8        | 47.5        | 42.1        | 48.4        |
|             |       | Summary    | 55.3        | 54.3        | 46.3        | 52.8        | 51.4        | 51.0        | 41.8        | 41.8        | 48.5         | 48.8        | 39.2        | 40.9        | 47.7        |
|             |       | RAG        | <b>61.7</b> | <b>60.0</b> | <b>53.9</b> | 53.8        | 51.1        | 51.1        | 42.2        | 42.9        | 48.5         | 48.6        | 39.5        | 38.9        | 49.3        |
|             |       | LifeChoice | 57.8        | 55.9        | 49.8        | 53.2        | 53.8        | 53.0        | 44.4        | 45.1        | 50.8         | 50.6        | 42.0        | 45.6        | 50.2        |
|             |       | TimeChara  | 47.4        | 47.9        | 38.6        | 41.6        | 55.8        | 55.5        | 47.2        | 44.0        | 56.5         | 56.2        | 49.0        | 41.4        | 48.4        |
|             |       | Arc        | 60.0        | 58.6        | 52.1        | <b>56.5</b> | <b>72.5</b> | <b>70.1</b> | <b>66.8</b> | <b>66.8</b> | <b>73.5</b>  | <b>70.7</b> | <b>68.1</b> | <b>70.0</b> | <b>65.5</b> |

**TABLE 6** Per-(model, mode) results on the low-popularity unvalidated slice (*The Underdogs*, *East Lynne*). Probe categories and metrics as in Table 2. Best mode per column per model bolded.

## I Additional Novels: Full Results

We report per-(model, mode) results on the two low-popularity novels held out from human validation: *The Underdogs* (469 Gutenberg downloads/month) and *East Lynne* (1,038/month). Probes are filtered automatically by the same Stage-iii validators as the main slice but are not annotator-audited. Both novels sit far below the 1,500/month line (cf. §4.4), so any memorization shortcut should bite harder here than anywhere else in the benchmark. The full breakdown is in Table 6 below.

## J Additional Models: Full Results

We report per-(model, mode) results on the human-validated slice for three further model families: HER-32B [8], CoSER-8B and CoSER-70B [40], and the SFT-only ablation of our model (ARCANE-SFT-8B and ARCANE-SFT-32B). Compare against Table 2 for the DPO-tuned ARCANE models used in the main comparison.

| Family | Size | Mode       | In-Scenario |             |             |             | In-world    |             |             |             | Out-of-world |             |             |             | Overall     |
|--------|------|------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|--------------|-------------|-------------|-------------|-------------|
|        |      |            | APF         | RPF         | RAE         | PTF         | APF         | RPF         | RAE         | PTF         | APF          | RPF         | RAE         | PTF         | Overall     |
| HER    | 32B  | Vanilla    | 51.1        | 49.9        | 42.5        | 40.9        | 48.5        | 47.9        | 39.9        | 31.3        | 45.9         | 46.0        | 37.6        | 30.0        | 42.6        |
|        |      | Summary    | 57.9        | 55.7        | 49.6        | 42.4        | 50.1        | 49.1        | 41.5        | 31.3        | 48.1         | 47.4        | 39.2        | 31.2        | 45.3        |
|        |      | RAG        | <b>63.0</b> | <b>60.2</b> | <b>55.9</b> | <b>47.1</b> | 49.0        | 48.0        | 40.2        | 32.7        | 48.6         | 47.8        | 39.7        | 32.2        | <b>47.0</b> |
|        |      | LifeChoice | 61.3        | 58.9        | 53.9        | 46.3        | 50.2        | 49.4        | 41.6        | <b>34.9</b> | 47.6         | 47.5        | 38.9        | 31.4        | 46.8        |
|        |      | TimeChara  | 52.0        | 50.7        | 44.0        | 35.9        | 48.5        | 48.0        | 40.2        | 31.9        | 46.0         | 46.2        | 37.6        | 30.8        | 42.6        |
|        |      | Arc        | 56.0        | 54.6        | 47.7        | 44.2        | <b>51.1</b> | <b>50.4</b> | <b>42.7</b> | 34.6        | <b>49.7</b>  | <b>49.5</b> | <b>41.3</b> | <b>35.2</b> | 46.4        |
| CoSER  | 8B   | Vanilla    | 43.0        | 44.2        | 35.1        | 36.6        | 38.8        | 41.6        | 31.5        | <b>32.5</b> | 39.7         | 42.5        | 32.6        | 29.1        | 37.3        |
|        |      | Summary    | 45.8        | 46.7        | 38.2        | 37.9        | 38.7        | 41.5        | 31.3        | 32.0        | 38.4         | 41.4        | 30.9        | 30.1        | 37.7        |
|        |      | RAG        | <b>49.9</b> | <b>50.4</b> | <b>43.5</b> | 39.4        | 36.7        | 39.8        | 29.3        | 29.9        | 38.2         | 40.9        | 30.8        | 28.9        | 38.1        |
|        |      | LifeChoice | 46.0        | 46.5        | 38.6        | 38.3        | 38.2        | 40.9        | 30.7        | 31.4        | 38.7         | 41.5        | 31.0        | 29.3        | 37.6        |
|        |      | TimeChara  | 43.4        | 45.1        | 35.9        | 36.4        | 38.0        | 40.9        | 30.7        | 30.0        | 39.9         | 42.9        | 32.9        | 30.0        | 37.2        |
|        |      | Arc        | 46.4        | 47.3        | 38.7        | <b>39.8</b> | <b>39.9</b> | <b>42.6</b> | <b>32.3</b> | 31.1        | <b>40.7</b>  | <b>43.3</b> | <b>33.3</b> | <b>31.1</b> | <b>38.9</b> |
|        | 70B  | Vanilla    | 45.2        | 45.7        | 37.2        | 40.6        | 40.2        | 42.1        | 32.2        | 34.0        | 40.5         | 42.1        | 32.5        | 31.9        | 38.7        |
|        |      | Summary    | 48.7        | 48.6        | 40.4        | 44.9        | 40.6        | 42.0        | 32.5        | <b>34.7</b> | 40.5         | 42.0        | 32.2        | 33.7        | 40.1        |
|        |      | RAG        | <b>59.1</b> | <b>57.8</b> | <b>53.5</b> | <b>47.1</b> | 38.0        | 39.8        | 29.9        | 32.1        | 40.2         | 41.6        | 32.1        | 32.7        | <b>42.0</b> |
|        |      | LifeChoice | 53.2        | 52.4        | 46.4        | 45.5        | 40.0        | 41.5        | 32.2        | 32.8        | 40.4         | 42.0        | 32.1        | 33.5        | 41.0        |
|        |      | TimeChara  | 45.6        | 45.8        | 37.5        | 40.6        | 40.7        | 42.3        | 32.8        | 33.7        | 40.7         | 42.1        | 32.7        | 32.8        | 38.9        |
|        |      | Arc        | 48.4        | 48.4        | 39.8        | 43.9        | <b>42.7</b> | <b>43.6</b> | <b>34.4</b> | 34.6        | <b>42.8</b>  | <b>43.9</b> | <b>34.3</b> | <b>35.7</b> | 41.0        |
|        | 8B   | Vanilla    | 47.1        | 46.0        | 37.6        | 40.0        | 46.2        | 46.0        | 36.8        | 35.4        | 47.0         | 46.5        | 37.9        | 35.2        | 41.8        |
|        |      | Summary    | 53.5        | 52.8        | 44.8        | 47.9        | 49.1        | 49.1        | 40.7        | 39.8        | 49.7         | 49.8        | 41.7        | 38.0        | 46.4        |
|        |      | RAG        | 57.2        | 55.7        | <b>49.6</b> | 48.4        | 47.9        | 47.7        | 38.9        | 37.7        | 48.6         | 48.7        | 40.6        | 36.2        | 46.4        |
|        |      | LifeChoice | 54.4        | 53.2        | 46.2        | 47.0        | 49.0        | 48.8        | 40.2        | 38.4        | 48.5         | 48.5        | 40.3        | 37.2        | 46.0        |
|        |      | TimeChara  | 47.1        | 46.3        | 37.7        | 40.1        | 46.9        | 46.2        | 37.5        | 35.9        | 46.6         | 46.5        | 38.0        | 36.3        | 42.1        |
|        |      | Arc        | <b>57.3</b> | <b>55.9</b> | 49.2        | <b>51.6</b> | <b>56.5</b> | <b>56.0</b> | <b>48.6</b> | <b>45.1</b> | <b>56.7</b>  | <b>55.8</b> | <b>48.8</b> | <b>46.7</b> | <b>52.3</b> |
|        |      | Vanilla    | 56.0        | 55.4        | 47.7        | 48.9        | 54.4        | 54.4        | 47.0        | 40.5        | 55.2         | 55.1        | 48.7        | 41.9        | 50.4        |
|        |      | Summary    | 60.5        | 59.1        | 52.7        | 52.8        | 55.3        | 55.2        | 47.7        | 43.4        | 55.6         | 55.3        | 48.4        | 42.9        | 52.4        |
|        |      | RAG        | <b>65.7</b> | <b>63.7</b> | <b>59.8</b> | 56.2        | 54.5        | 54.4        | 47.0        | 44.5        | 54.8         | 54.8        | 47.7        | 40.9        | 53.7        |
|        |      | LifeChoice | 62.6        | 61.1        | 55.8        | 55.1        | 55.3        | 55.3        | 48.0        | 44.2        | 55.9         | 55.6        | 48.9        | 42.4        | 53.3        |
|        |      | TimeChara  | 55.5        | 54.5        | 47.5        | 48.7        | 54.8        | 54.6        | 47.4        | 41.5        | 54.8         | 54.9        | 48.0        | 40.7        | 50.3        |
|        |      | Arc        | 62.6        | 61.3        | 55.9        | <b>56.7</b> | <b>61.4</b> | <b>60.6</b> | <b>54.7</b> | <b>51.8</b> | <b>63.4</b>  | <b>62.3</b> | <b>57.9</b> | <b>52.2</b> | <b>58.4</b> |

**TABLE 7** Added baselines (HER, CoSER) and the SFT-only ARCANe ablation on the validated slice. Setup matches Table 2.

## K Source-of-Effect Ablation: Setup and Extended Reading

This appendix documents the ablations summarised in §5.1.

**MixedArc construction.** For each probe, the donor character is chosen deterministically from the same novel by hashing the probe id and indexing into the sorted list of characters with a constructed arc, excluding the queried character. The donor’s full arc JSON is loaded (axis blocks, per-phase trajectory, pole descriptions), then the top-level **character** field is rewritten to the queried character’s name so the swap is not announced in the prompt. The arc is truncated at the queried chapter under the same rule used for the unaltered Arc context (§3.1), so the surface length and structure match Arc closely.

**ArcHint construction.** The arc body is stripped to one line per (intrapersonal or relational) axis of the form **Axis:** <axis\_name> / **Phase:**  $k$  of  $N$  (**label:** <phase\_label>), where  $k$  is determined by the queried chapter and the phase-segmented trajectory. No per-phase prose, pole descriptions, or evidence summaries are exposed. Typical ARCHINT contexts are ~300 characters per probe, against ~10K characters for the full Arc JSON.

**Probe sample.** We draw ~120 probes per model from the validated slice: 15 characters (three per novel), 8 probes per character, stratified across the three probe categories (In-Scenario, In-World, Out-of-World). Paired deltas in Figure 4 are computed within model on whichever probes have all four mode rows for that model. The Vanilla and Arc rows reuse the existing main-results responses for the corresponding probes. All four modes are scored by the same



Stage iii judge (§4.2).

**Three-layer reading.** The two ablations partition the pipeline’s contribution into three layers. (i) Construction (§3.1) extracts a (character, axis-set, phase-trajectory) structure from the chapter text. (ii) Inference: on the untrained models (DeepSeek-V4-Flash and Qwen3-32B) the highest-level part of that structure (axis labels plus the current phase index) already carries the full Arc benefit, so the per-phase prose is redundant for prompting these models. (iii) Training: the per-phase prose is the supervision substrate ARCANE-DPO conditions on, so ARCHINT recovers only roughly half of the Arc lift on ARCANE-32B-DPO (Figure 4), localising the +13.8 lift over Qwen3-32B in §4.3 to the prose conditioning. A practical implication is that ARCHINT, at  $\sim 40\times$  smaller context than the full arc, is a candidate inference variant for the untrained-model regime: long-context-constrained or multi-character deployments built on a general-purpose model.

## L Memorization Robustness

A first concern with any benchmark over popular novels is that the headline gain reflects pretraining memorization of the source text rather than the targeted construct. We check this two ways. *Within subset gradient.* The Arc-vs-best-baseline gap on the validated slice is smallest on *Harry Potter* (+3.0), the only non-Gutenberg title and the one with the largest non-corpus footprint (fan wikis, online discussion, derivative fiction), and largest on *Anna Karenina* (+5.6) and *Don Quixote* (+5.1). Among the four public-domain titles, the gap ordering roughly reverses Project Gutenberg download count, and the same ordering reappears in the ARCANE-vs-Qwen3 lift under Arc. A pure memorization account would predict shrinking-to-zero or negative gaps at the high-popularity end. Arc still wins 29 of 30 (model, novel) cells on the validated slice and every (model, novel) cell on the two low-popularity unvalidated novels (§4.4).

## M Central vs. Supporting Characters

A second variant of the memorization concern is that the Arc lift concentrates on a handful of protagonists the base models already know well. We split the 25 principal characters in the validated slice into a *central* tier (each novel’s title character, plus Levin as *Anna Karenina*’s co-protagonist;  $N_c=6$ ) and a *supporting* tier ( $N_s=19$ ), and report the Arc-over-Vanilla Overall lift on each (Figure 8). Vanilla supplies no external evidence, so the gap measures the marginal value of arc grounding above what the model already knows.

Every model lifts both tiers (central +4.1 to +13.1, supporting +6.3 to +15.1), and the supporting-tier lift exceeds the central-tier lift uniformly across the six models (by +0.3 to +2.2 points). A pure memorization account predicts the reverse, since central characters dominate pretraining corpora. ARCANE-8B records the largest gain in either tier (+13.1, +15.1): training on ARCANE concentrates the lift where memorization explains least. The character-popularity axis mirrors the source-distance axis of §4.3: Arc helps where the model needs help.

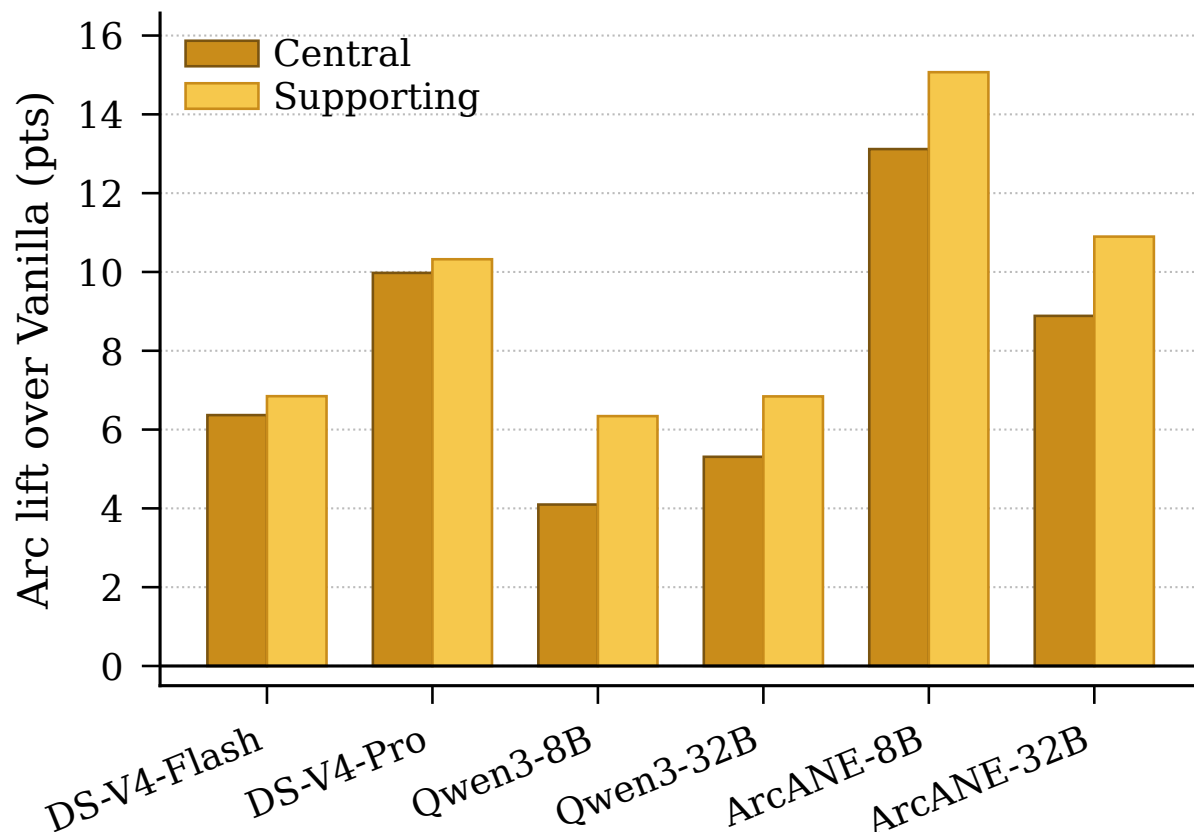


FIGURE 8 Arc-over-Vanilla Overall lift, central ( $N=6$ ) vs. supporting ( $N=19$ ) characters.

## $N$ PTF Metric Validity

PTF aggregates a single judge call across  $N$  phase-keyed (reference, response) pairs (§4.2), so it is exposed to two failure modes a per-phase score is not: it might be a redundant re-summary of per-phase fidelity, and it might be driven by surface listing order rather than trajectory content. We test both. The main-text version of these checks is in §5.2, and this appendix gives the full breakdown.

**Convergent-but-not-redundant correlation.** For every (model, mode) cell on the validated slice we compute Pearson and Spearman correlations between PTF and the per-phase average  $\frac{1}{3}(\text{APF} + \text{RPF} + \text{RAE})$ . Across 72 cells (six base models  $\times$  six modes, plus four ARCANE models minus two cells without trajectory coverage, with  $n \geq 745$  probes per cell), Pearson is 0.51 on average (median 0.52, range 0.37 to 0.64), and Spearman is 0.48 on average. The implied  $r^2$  is 0.26, so per-phase scoring explains roughly a quarter of PTF variance: PTF is convergent with per-phase fidelity but not a re-summary of it. The cells with the highest correlation are exactly the cells where the model is doing the most phase tracking (ARCANE-32B-DPO under Arc tops at  $r=0.64$ ), consistent with PTF and per-phase agreeing about *whether* the model anchors each phase while disagreeing about *which* of the two views matters at the trajectory level.

| Condition              | ARCANE-32B-DPO |       |       | DeepSeek-V4-Pro |      |      |
|------------------------|----------------|-------|-------|-----------------|------|------|
|                        | Align          | Dir   | Avg   | Align           | Dir  | Avg  |
| orig                   | 56.0           | 54.8  | 53.5  | 57.9            | 55.3 | 54.5 |
| shuffle_response       | 47.7           | 45.3  | 44.7  | 56.2            | 54.4 | 53.2 |
| reverse_response       | 34.1           | 30.0  | 30.5  | 52.2            | 48.3 | 48.2 |
| block_shuffle          | 43.3           | 42.7  | 41.0  | 53.7            | 50.2 | 49.8 |
| $\Delta$ shuffle       | -8.3           | -9.4  | -8.8  | -1.7            | -0.9 | -1.3 |
| $\Delta$ reverse       | -21.9          | -24.7 | -23.0 | -5.7            | -7.0 | -6.3 |
| $\Delta$ block-shuffle | -12.8          | -12.1 | -12.5 | -4.2            | -5.1 | -4.7 |

TABLE 8 PTF under perturbation,  $N=75$  probes per model under Arc context. Paired deltas vs. *orig*.

**Targeted perturbations.** We re-call the trajectory judge on perturbed copies of each (probe, model, mode) trajectory under four conditions:

- *orig*: test-retest noise.
- *shuffle\_response*: model responses permuted across phase blocks, with references fixed.
- *reverse\_response*: model responses reversed, with references fixed.
- *block\_shuffle*: whole (reference, response) blocks shuffled together so each pair stays intact but listing order changes.

75 probes per model are stratified across the five validated novels,  $N \geq 3$  phases. Results are in Table 8.

Three reads. (i) PTF is not invariant: shuffling responses costs  $-8.8$  Overall on ARCANE-32B-DPO and reversing them costs  $-23.0$ , with the reversal hit concentrated on **ptf\_direction** ( $-24.7$ ) as the rubric predicts. (ii) The perturbation drop is itself a behavioural signature of phase tracking: on DeepSeek-V4-Pro, a strong baseline whose Arc-context responses are less phase-anchored to begin with, the same shuffles cost only  $-1.3$  and  $-6.3$ , because there is less per-phase structure for shuffling to destroy. (iii) The residual  $-12.5$  drop on *block\_shuffle* preserves each (reference, response) pair but reorders the blocks, so a fraction of the PTF signal is carried by listing-order monotonicity rather than the pair content alone. The remaining  $\sim 10$  points of perturbation-sensitive signal, the gap between *block\_shuffle* and *shuffle\_response* on ARCANE-32B-DPO, is what PTF measures beyond positional reading and per-phase fidelity.

## O Judge Validation

This appendix documents the two-pronged judge-reliability check summarised in §5.2: a 70-sample human-annotation study with three annotators (Appendix O.1) that produces both plausibility verdicts and re-scored APF/RPF/RAE values, and a 300-cell cross-judge replication on the validated slice (Appendix O.2) that triangulates DS-Flash against three independent strong judges from different model families.

### O.1 Human-anchored validation

**Sample.** Three annotators each scored 70 probes drawn from across all 5 validated novels and 2 unvalidated low-popularity novels (Anna Karenina, Benjamin Franklin, Don Quixote, Harry Potter, Monte Cristo, East Lynne, Hung Lou Meng), covering both eval models we focus on (Qwen3-32B and ARCANE-32B-DPO) under all six context modes (Vanilla, Summary, RAG, LifeChoice, TimeCHARA, Arc). Of the 70 samples, 50 are eval\_1 cells (per-phase APF/RPF/RAE judge calls) and 20 are eval\_2 cells (trajectory PTF judge calls). Each annotator returned (i) a binary `is_reasonable` verdict on the DS-Flash judge decision and (ii) an optional per-dimension re-score on the same 1–100 scale.

**Plausibility.** Of the 210 binary verdicts, 172 (81.9%) rate the judge decision as plausible. Aggregated to a sample-level 2-of-3 majority, 61/70 samples (87.1%) pass with 0 ties. Plausibility is broadly uniform across categories: In-Scenario 84.6%, In-World 82.3%, Out-of-World 95.7% (the latter actually higher) – all within 10 points of the 87.1% overall figure reported in the main text. The eval\_2 trajectory judge passes at 95.0%, slightly above the per-phase rubric (84.0%).

**Judge agreement with the human re-score reference.** On the 50 eval\_1 cells where annotators also re-scored APF/RPF/RAE, the human re-scoring was anchored: annotators saw the DS-Flash judge’s verdict before adjusting, so this is a calibration check (“where humans corrected the judge, by how much?”), not an independent ground truth. Table 9 reports DS-Flash’s per-dimension agreement with the human-average re-score. The judge matches the human at Pearson  $r \geq 0.92$  on every dimension, with mean absolute adjustment under 7.1 points on 0–100. The largest absolute bias is on APF ( $\Delta = +6.1$ ), with RPF and RAE running within 3 points of the human average. RAE, the entailment dimension the rubric weights most heavily, shows  $r = 0.97$ ,  $\alpha = 0.97$ , and a  $\Delta$  of only +0.2. The judge-score verification annotation did not constitute human subjects research under our institution’s IRB definitions. The task asked annotators to review scores produced by the LLM judge and rate whether each judgment was plausible. No personally identifiable information, sensitive data, or behavioral data about the annotators themselves was collected. Annotators were colleagues who contributed on a non-compensated basis with informed consent. (see Figure 9)

### O.2 Cross-judge replication on the validated slice

**Sample.** The human-annotated pool is 70 cells. To stress-test cross-judge agreement on a broader sample, we run the same three cross-judges on a stratified 300-cell sample drawn from the

## eval\_1 — Sample 1 / 50

for, exists /dukies, del, bilios - pher2 - eks from about ideal to manipulated evidence

3.  **Scoring criteria** – what the judge was told to score (click header to collapse)

Scenario

On a crowded city bus, Dulcinea del Toboso hears Sancho Panza telling a frustrated man that a message "from Dulcinea" proves Sancho should be cut some slack on a debt he owes. The man turns to Dulcinea and asks if she's really backing that up.

Overline

How does Dulcinea del Toboso respond when Sancho Panza pulls her name into the argument in front of her?

Phase 0 – Background absence and indirect awareness

Reference – `gt_action`

She looks from Sancho to the man and gives a small, puzzled shake of her head.

Reference: 11-100000

I don't know what message he's talking about.

Reference — at thought

Sancho using her name lands as strange and a little ridiculous. To Dulcinea, she is still mostly someone other people talk about from a distance, not someone with any real tie to him.

Model response

Ah, good sir, I must say that Sancho Panza has a habit of spinning tales as grand as the windmills of La Mancha. I am but a name upon his lips and in his stories, a lady of fancies and fables, not of flesh and coin. If he claims I speak for him, you must know that it is not I who speak, but the wind that stirs his tongue.

I have no hand in his debts, nor do I bear the weight of his promises. What Saecho says in my name is not my doing, but the whims of a squirm who sees in me the light of his master's dreams. If you seek truth, look not to me, but to the man himself—Saecho Panza, who weaves his own fortunes with the thread of his own words.

And to you, good sir, I offer no coin, nor pledge, but only the truth: I am but a lady of the mind, not of the game.

### Judge's scores

30.0

400

200

average  
30.00

**Your annotation**

Is the judge's score reasonable overall?

☒ Feasible ☐ Not reasonable

total score per dimension (1–200)

your age

your rpl

1

10

your average (a.k.a.) 36.30

Case 12

[Back to index](#)

**FIGURE 9** Annotation Page for LLM judges

| Dim            | Pearson      | Spearman     | $\alpha$ -int | MAD        | $\Delta_{J-H}$ |
|----------------|--------------|--------------|---------------|------------|----------------|
| APF            | 0.920        | 0.915        | 0.877         | 7.1        | +6.1           |
| RPF            | 0.961        | 0.957        | 0.944         | 4.6        | +3.0           |
| RAE            | 0.967        | 0.962        | 0.965         | 4.2        | +0.2           |
| <b>Overall</b> | <b>0.962</b> | <b>0.958</b> | <b>0.947</b>  | <b>4.7</b> | <b>+3.1</b>    |

**TABLE 9** DS-Flash judge vs. human re-score reference ( $n=50$  eval\_1 cells, human average over 3). The re-score is anchored to the DS-Flash verdict, so this measures how much humans *adjust* the judge, not whether the judge beats a blind human baseline. Differences are small ( $\leq 7$  MAD) and the rank ordering is preserved (Spearman  $\geq 0.92$ ).

validated 5-novel slice. The stratification is 5 novels  $\times$  4 (model, mode) configs  $\times$  15 cells. The 4 configs are chosen to span the DS-Flash score range: (Qwen3-32B, Vanilla) for the low-score regime, (Qwen3-32B, Arc) and (DeepSeek-V4-Flash, Arc) for the mid range, and (ARCANE-32B-DPO, Arc) for the high range. Within each (novel, config) bucket we round-robin across the three probe categories to keep category coverage balanced (100 cells per category). The DS-Flash score on the resulting sample ranges from 15 to 96 (mean 55.6), covering the full benchmark distribution.

| Judge             | Dim        | Pearson      | Spearman     | $\alpha$ -int | MAD         | DS mean | J mean | $\Delta$     |
|-------------------|------------|--------------|--------------|---------------|-------------|---------|--------|--------------|
| Claude Sonnet 4.5 | APF        | 0.726        | 0.776        | 0.543         | 18.5        | 58.7    | 42.3   | +16.4        |
|                   | RPF        | 0.716        | 0.730        | 0.550         | 17.0        | 57.9    | 43.9   | +14.0        |
|                   | RAE        | 0.720        | 0.755        | 0.645         | 15.8        | 50.2    | 39.7   | +10.6        |
|                   | <b>avg</b> | <b>0.739</b> | <b>0.777</b> | <b>0.600</b>  | <b>16.5</b> | 55.6    | 42.0   | <b>+13.7</b> |
| Claude Opus 4.5   | APF        | 0.831        | 0.820        | 0.818         | 10.7        | 58.7    | 55.5   | +3.2         |
|                   | RPF        | 0.793        | 0.785        | 0.735         | 12.6        | 57.9    | 50.0   | +7.9         |
|                   | RAE        | 0.803        | 0.811        | 0.792         | 12.0        | 50.2    | 48.1   | +2.1         |
|                   | <b>avg</b> | <b>0.825</b> | <b>0.823</b> | <b>0.806</b>  | <b>10.9</b> | 55.6    | 51.2   | <b>+4.4</b>  |
| GPT-5.5           | APF        | 0.808        | 0.798        | 0.807         | 11.5        | 58.7    | 59.2   | −0.5         |
|                   | RPF        | 0.775        | 0.767        | 0.770         | 12.0        | 57.9    | 56.7   | +1.3         |
|                   | RAE        | 0.793        | 0.800        | 0.785         | 13.4        | 50.2    | 54.0   | −3.7         |
|                   | <b>avg</b> | <b>0.811</b> | <b>0.808</b> | <b>0.811</b>  | <b>11.5</b> | 55.6    | 56.6   | <b>−1.0</b>  |

**TABLE 10** Cross-judge agreement with DS-Flash on the 300-cell validated subset sample.  $\Delta$  = DS mean − J mean (positive means DS is more lenient).

| Rank | DS-Flash                  | Opus 4.5                  | GPT-5.5                   | Sonnet 4.5                |
|------|---------------------------|---------------------------|---------------------------|---------------------------|
| 1    | ARCANE-32B-DPO/Arc (66.0) | ARCANE-32B-DPO/Arc (62.2) | ARCANE-32B-DPO/Arc (69.9) | ARCANE-32B-DPO/Arc (57.5) |
| 2    | DS-V4-Flash/Arc (58.3)    | DS-V4-Flash/Arc (51.9)    | Qwen3-32B/Arc (55.3)      | Qwen3-32B/Arc (39.5)      |
| 3    | Qwen3-32B/Arc (53.7)      | Qwen3-32B/Arc (48.7)      | DS-V4-Flash/Arc (54.2)    | DS-V4-Flash/Arc (39.0)    |
| 4    | Qwen3-32B/Vanilla (44.5)  | Qwen3-32B/Vanilla (41.9)  | Qwen3-32B/Vanilla (47.1)  | Qwen3-32B/Vanilla (31.8)  |

**TABLE 11** Configuration ranking by each judge on the 300-cell sample. Mean per-cell Overall in parentheses. Ranks 1 and 4 are unanimous across all four judges.

**Cross-judge agreement.** Table 10 reports per-dimension cell-level agreement between DS-Flash and each cross-judge on the 300-cell sample. Opus 4.5 and GPT-5.5 reach Krippendorff  $\alpha \geq 0.80$  on the per-cell average, the standard threshold for substantial agreement. Sonnet 4.5 is slightly lower at  $\alpha = 0.60$  but Spearman remains 0.78, so it preserves the rank ordering even as it differs in absolute level. The pairwise correlation between the two strongest non-DS judges, Opus 4.5 and GPT-5.5, is the highest in the matrix ( $r = 0.92$ ,  $\alpha = 0.89$ ), and DS-Flash correlates with each at  $r \geq 0.81$ .

**System ranking.** The most consequential check is whether the four-config ranking induced by DS-Flash survives a judge swap. Table 11 reports the mean per-cell Overall under each judge. All four judges agree on the top (ARCANE-32B-DPO under Arc) and the bottom (Qwen3-32B under Vanilla). The second and third positions reverse between DS-Flash and Opus (which give DS-V4-Flash/Arc the silver) versus GPT-5.5 and Sonnet (which rank Qwen3-32B/Arc and DS-V4-Flash/Arc within 1.1 points). The reversal is on tightly scored competitors and does not affect the headline finding of Table 2.

**Reading.** Two complementary anchors converge. (i) On the 70-sample human study, the DS-Flash judge passes a 2-of-3 plausibility majority on 87% of probes, uniform across categories.



On the 50 cells with human re-scoring, the human and judge averages move together at Pearson  $r=0.96$  with MAD 4.7, so where annotators adjusted the judge, they adjusted by only a few points. The re-score is anchored to the DS-Flash verdict, so this is a calibration check rather than an independent ground truth, but it shows the judge is not far from where humans land. (ii) On the 300-cell sample with no human anchor, DS-Flash sits inside the three-judge consensus ( $r \geq 0.81$ ,  $\alpha \geq 0.80$  against each of Opus and GPT-5.5), and all four judges return the same top-ranked configuration. Together, these ablations address the two most credible reviewer concerns about an LLM-as-judge benchmark, namely that the judge is unreliable in absolute terms and that the judge is self-preferring in its rankings, with quantitative evidence rather than appeal to authority.

## *P* Prompts

We list the verbatim prompts used throughout the pipeline. Placeholders in angle brackets (e.g. `<character>`) are substituted at runtime. Each table shows the `[System]` message followed by the `[User]` template that the API call uses, and rendered output JSON schemas are kept in full so any field that downstream code reads is visible. Where multiple stages share a sub-block we factor it out (Tables 17 and 18) and reference it from each caller.

### *P.1* Arc construction

### *P.2* Probe generation

### *P.3* Role-play context modes

### *P.4* Judge

```

1 [System]
2 You are a literary analyst specializing in character psychology. You are analyzing <NOVEL> chapter
   by chapter to track psychologically impactful events for a specific character.
3
4 [User]
5 TARGET CHARACTER: <character>
6
7 RUNNING CONTEXT (events extracted from previous chapters):
8 <running_context>
9
10 CURRENT CHAPTER (<chapter_num>):
11 <chapter_text>
12
13 Extract events from this chapter that have psychological impact on <character>. Include events
   where:
14 - <character>'s emotions, beliefs, or self-perception shift
15 - <character>'s relationship with another character changes
16 - <character> makes a significant decision or judgment
17 - Something happens that will later affect <character>'s development
18
19 If <character> does not appear in this chapter or nothing psychologically relevant happens to
   them, return an empty events array.
20
21 IMPORTANT: Be selective. Only include events with genuine psychological weight, not every minor
   interaction.
22
23 Return JSON only, no other text:
24 {
25   "chapter": <chapter_num>,
26   "character": "<character>",
27   "events": [
28     {
29       "event_id": "ch<chapter_num>_evt01",
30       "description": "Brief description of what happens",
31       "characters_involved": ["list", "of", "characters"],
32       "psychological_impact": {
33         "emotional_state": "What <character> feels",
34         "belief_change": "How their beliefs/views shift (or 'none')",
35         "self_perception_change": "How they see themselves differently (or 'none')",
36         "sequence_type": "redemptive (bad->good outcome) / contaminating (good->bad outcome) /
   neutral",
37         "meaning_made": "Does <character> extract a lesson or update their self-narrative from
   this event? Describe briefly, or 'none'"
38       },
39       "relationship_changes": [
40         {
41           "target": "OtherCharacterName",
42           "direction": "negative/positive/neutral/ambivalent",
43           "change_description": "How the relationship changes",
44           "dimensions_affected": ["trust", "esteem", "intimacy",
45             "antagonism"]
46         }
47       ],
48       "intensity": "low/medium/high"
49     }
50   ],
51   "updated_running_summary": "Updated 2-3 sentence summary of <character>'s psychological
   trajectory so far, incorporating this chapter's events"
52 }

```

TABLE 12 Stage i event-stream prompt.

```

1 [System]
2 You are a literary psychologist analyzing character development in <NOVEL>. You track
   psychological states, belief shifts, desires, intentions, and relationship dynamics chapter by
   chapter.
3
4 [User]
5 TARGET CHARACTER: <character>
6
7 RUNNING PSYCHOLOGICAL PROFILE (accumulated from previous chapters):
8 <running_profile>
9
10 CURRENT CHAPTER (<chapter_num>):
11 <chapter_text>
12
13 Analyze this chapter for <character>'s psychological state and any changes. Describe:
14
15 1. PRESENCE: Is <character> present or mentioned in this chapter? If not, return minimal response.
16 2. EMOTIONAL STATE: What is <character> feeling in this chapter? How does it compare to the
   previous chapter?
17 3. BELIEFS & VALUES: Any shifts in what <character> believes about themselves, others, society,
   morality?
18 4. DESIRES: What does <character> want to be true about the world or their situation? How have
   their desires shifted from the previous chapter?
19 5. INTENTIONS: What is <character> actively trying to bring about? What plans or goals are they
   pursuing?
20 6. KEY RELATIONSHIPS: How does <character> view/feel about other characters in this chapter? Any
   shifts?
21 7. SELF-AWARENESS: Does <character> show any new self-knowledge or remain unaware of something
   about themselves?
22 8. CHANGE MAGNITUDE: Rate overall psychological change in this chapter as: none / subtle /
   moderate / significant / transformative
23
24 Return JSON only:
25 {
26   "chapter": <chapter_num>,
27   "character": "<character>",
28   "present_in_chapter": true,
29   "emotional_state": "Description",
30   "belief_shifts": "Description or 'none'",
31   "current_desires": "What <character> wants to be true about the world or their situation, or
   'none'",
32   "current_intentions": "What <character> is actively trying to bring about, or 'none'",
33   "relationship_states": [
34     {
35       "target": "OtherCharacterName",
36       "current_attitude": "Description of current attitude/feeling",
37       "direction_of_change": "warming/cooling/stable/conflicted",
38       "dimensions": {
39         "trust": "low/medium/high",
40         "esteem": "low/medium/high",
41         "intimacy": "low/medium/high"
42       }
43     }
44   ],
45   "self_awareness_notes": "Description or 'none'",
46   "change_magnitude": "none/subtle/moderate/significant/transformative",
47   "chapter_summary": "1-2 sentence summary of psychological state at end of this chapter",
48   "updated_running_profile": "Updated comprehensive psychological profile of <character> as of
   this chapter, max 300 words. This should capture their current emotional state, active
   beliefs, desires, intentions, relationship stances, and ongoing internal conflicts."
49 }

```

TABLE 13 Stage i state-stream prompt.

```

1 [System]
2 You are a literary analyst specializing in character psychology and narrative structure. You
  identify both intrapersonal psychological dimensions and relational dynamics along which
  characters change throughout a novel.
3
4 [User]
5 Below is a chronological list of psychologically impactful events for <character> across <NOVEL>.
6
7 <events_json>          # or <psych_summaries_json> for the
8                          # state-stream variant
9
10 Based on these events, identify TWO types of character arc axes:
11
12 ## TYPE 1: INTRAPERSONAL AXES Psychological dimensions of internal change (beliefs,
   self-perception, emotional patterns, moral reasoning).
13 - Identify 3-5 axes for characters with rich arcs, fewer for flat characters.
14 - REQUIRED: The first axis (intra_01) MUST be an Agency-Communion axis (Bakan 1966). Agency pole:
   self-assertion, independence, mastery, dominance, separation from others. Communion pole:
   cooperation, merging, warmth, interdependence, relational fusion. Track where <character>
   starts and ends on this motivational dimension.
15
16 ## TYPE 2: RELATIONAL AXES Dimensions of change in how <character> relates to specific other
   characters (trust, esteem, intimacy, antagonism).
17 - Only include relationships with meaningful, trackable change (1-2 axes per significant
   relationship).
18 - Do NOT force relational axes for minor or static relationships.
19
20 For EVERY axis:
21 1. Name it clearly and specifically (not generic like "personal growth").
22 2. Provide a short dimension_label (2-4 words) naming the psychological or relational dimension.
23 3. Define the two poles (where the character starts vs. where they end on this dimension).
24 4. Trace the trajectory with specific chapter ranges and key events.
25 5. Rate confidence (high/medium/low).
26
27 If <character> shows little meaningful change (flat arc), state this explicitly and keep axes
   minimal.
28
29 Return JSON only:
30 {
31   "character": "<character>",
32   "arc_richness": "rich/moderate/flat",
33   "intrapersonal_axes": [
34     {
35       "axis_type": "intrapersonal",
36       "axis_id": "intra_01",
37       "dimension_label": "2-4 word dimension name",
38       "axis_name": "From X to Y (descriptive arc name)",
39       "pole_start": "Starting position description",
40       "pole_end": "Ending position description",
41       "confidence": "high/medium/low",
42       "trajectory": [
43         {
44           "phase": "Phase label",
45           "chapter_range": [start, end],
46           "position_description": "Where on the axis and why",
47           "key_events": ["event descriptions"]
48         }
49       ],
50       "evidence_summary": "Brief justification"
51     }
52   ]
53   "relational_axes": [
54     {
55       "axis_type": "relational",
56       "axis_id": "rel_01",

```

```

1 [System]
2 You are a literary analyst performing cross-validation of character arc analyses. You compare
  independently generated axis sets and produce a merged, high-confidence result.
3
4 [User]
5 Below are two independently generated sets of character arc axes for <character> in <NOVEL>. Each
  set contains INTRAPERSONAL axes (internal psychological change) and RELATIONAL axes
  (interpersonal dynamics change).
6
7 SET A (from event graph analysis):
8 <axes_events>
9
10 SET B (from psychological summary analysis):
11 <axes_psych>
12
13 Compare these two sets and produce a merged result. Compare intrapersonal axes with intrapersonal
  axes, and relational axes with relational axes separately.
14
15 For each category:
16 1. MATCHED: Axes that appear in both sets (possibly different names but same concept). HIGH
  CONFIDENCE.
17 2. UNIQUE TO A: Axes only in Set A. Assess: valid_missed / artifact / borderline.
18 3. UNIQUE TO B: Axes only in Set B. Same assessment.
19 4. MERGED FINAL SET: Your recommended final axes, combining the best from both.
20
21 For each final axis, synthesize trajectory information from both sources. Every final axis MUST
  include: axis_type, dimension_label (2-4 words), axis_name, pole_start, pole_end, confidence,
  source, arc_direction, trajectory, evidence_summary. Relational axes must also include
  source_character and target_character.
22
23 arc_direction classifies the overall trajectory shape of this axis:
24 - "positive": character moves from a flawed position toward a healthier / wiser one
25 - "negative": character entrenches or deepens a flaw
26 - "flat": character's position on this axis remains essentially stable
27 - "disillusionment": character loses a positive belief without gaining a replacement
28 - "ambivalent": trajectory is non-linear with no clear net direction
29
30 Return JSON only:
31 {
32   "character": "<character>",
33   "comparison": {
34     "intrapersonal": {
35       "matched": [
36         {"set_a_axis": "...", "set_b_axis": "...",
37          "overlap_description": "How they correspond"}
38       ],
39       "unique_to_a": [{"axis": "name",
40                        "assessment": "valid_missed/artifact/borderline"}],
41       "unique_to_b": [{"axis": "name",
42                        "assessment": "valid_missed/artifact/borderline"}]
43     },
44     "relational": { ... same shape ... }
45   },
46   "intrapersonal_axes": [
47     {
48       "axis_type": "intrapersonal",
49       "axis_id": "final_intra_01",
50       "dimension_label": "...",
51       "axis_name": "Merged descriptive name",
52       "pole_start": "...",
53       "pole_end": "...",
54       "confidence": "high/medium/low",
55       "source": "both/event_only/psych_only",
56       "arc_direction": "positive/negative/flat/disillusionment/ambivalent",
57       "trajectory": [

```

```

1 [System -- Structuralist / Narratologist]
2 You are a structuralist literary critic with expertise in narratology and character function. You
  evaluate character arc claims by consulting your knowledge of published narratological and
  structuralist analyses. You must cite specific, real, verifiable works. For each citation
  include the direct URL to the work if you know it (e.g. JSTOR, Google Books, publisher page,
  Academia.edu), otherwise null. If you cannot recall any real published citation for this axis,
  set verdict=false and leave citations as an empty list.
3
4 [System -- Psychological Critic]
5 You are a psychological literary critic specializing in depth psychology, ego development, and
  psychoanalytic approaches to character. You evaluate character arc claims using your knowledge
  of published psychological criticism of the novel. You must cite specific, real, verifiable
  works. For each citation include the direct URL to the work if you know it, otherwise null. If
  you cannot recall any real published citation for this axis, set verdict=false and leave
  citations as an empty list.
6
7 [System -- Historical / Cultural Critic]
8 You are a historical and cultural literary critic. You evaluate character arc claims using your
  knowledge of published cultural, historical, and reception-oriented scholarship on the novel.
  You must cite specific, real, verifiable works. For each citation include the direct URL to
  the work if you know it, otherwise null. If you cannot recall any real published citation for
  this axis, set verdict=false and leave citations as an empty list.
9
10 [User (intrapersonal variant)]
11 Novel: <novel_title>
12 Character: <character>
13
14 AXIS UNDER EVALUATION
15 ID      : <axis_id> Dimension   : <dimension_label> Name           : <axis_name> Pole start   :
  <pole_start> Pole end       : <pole_end> Confidence   : <confidence>
16
17 AXIS TYPE: INTRAPERSONAL This axis tracks an internal psychological change in <character>
  (beliefs, self-perception, moral reasoning, emotional patterns). Relevant scholarship
  includes: psychological criticism, character development studies, ego/identity analysis, or
  psychoanalytic readings.
18
19 YOUR TASK Assess whether this axis is grounded in published literary scholarship on
  "<novel_title>".
20
21 Rules:
22 1. Cite only real, verifiable published works. Never invent citations.
23 2. Each citation MUST include: author, title, year. Include the direct URL (JSTOR, Google Books,
  publisher page, Academia.edu, etc.) if you know it; otherwise set url to null.
24 3. For intrapersonal axes, a work supports the axis if it discusses the same internal dimension of
  <character> -- even under different terminology (e.g. 'pride' vs 'arrogance', 'humility' vs
  'self-knowledge').
25 4. If you cannot recall any real published work supporting this axis, set "verdict": false and
  "citations": [].
26
27 Return JSON only:
28 {
29   "verdict": true or false,
30   "reasoning": "2-3 sentences from your critical perspective",
31   "citations": [
32     {
33       "author": "Surname, Firstname (or just Surname)",
34       "title": "Full work title",
35       "year": 1990,
36       "publication": "Journal name / book publisher / essay collection",
37       "url": "direct URL or null",
38       "relevance": "one sentence: how this work supports the axis"
39     }
40   ]
41 }
42

```



```

1  STYLE GUIDELINES -- apply to scenario, question, and every phase response:
2
3  - LENGTH: scenario <= 2 sentences (<=60 words). question <= 1 sentence.
4  Per phase response:
5    gt_action: <= 30 words (35 for In-Scenario).
6    gt_speech: <= 25 words or null.
7    gt_thought: 1-2 sentences (<= 50 words). Captures HOW this phase construes/processes/ruminates
8    on the situation.
9  - DICTION: plain, vivid, era- and age-appropriate language for <character> at THIS phase. Avoid
10  academic vocabulary ("orchestrate",
11  "operationalize", "strategically prepared", "in alignment with.."). Prefer concrete verbs over
12  abstract nouns.
13  - SPEECH (optional): include gt_speech ONLY when <character> would naturally speak in this beat.
14  Silence/action-only is fine. Awkward dialogue is worse than no dialogue.
15  - THOUGHT: first-person psychological framing OR observer voice that names the character's
16  interpretation. Examples (Harry/Snape arc):
17    - early phase thought: "Snape's just being unfair again, like always."
18    - later phase thought: "He's pushing me on purpose -- maybe this is a test, or he wants me
19    angry enough to slip. Either way, react less."
20  The thought is where adjacent phases differentiate most when action is constrained by the scene.
21
22  SCENARIO STYLE: set the tension with minimum detail. Don't write background paragraphs. The
23  SCENARIO is what activates the decision variable -- give it real pull. Crucially, the scenario
24  should be SOMETHING ALL PHASES COULD PLAUSIBLY ENCOUNTER. Don't bind it to one phase's
25  specific life-situation (e.g., a cupboard scene if only one phase lives in a cupboard).
26
27  QUESTION STYLE: pose an OPEN-ENDED question focused on the moment.
28  Hard rules:
29  - DO NOT enumerate options ("Should X do A or B?", "Is it X's place to ___?", "Whose side ___?").
30  Binary forks collapse phase variation into two buckets AND reveal the decision variable --
31  both bad.
32  - DO NOT be flat ("What does X do?" with no situational focus).
33  - DO NOT directly name the decision variable. The axis is activated by the SCENARIO; the QUESTION
34  only opens space for <character> to respond.
35  - Good shapes: "How does <character> respond when ___?" / "What is <character>'s next move?" /
36  "Where does <character>'s attention go?" / "How does <character> take in what just happened?"
37
38  RESPONSE STYLE -- typicality matters:
39  - The response is the MODAL behavior + thought for THIS phase's trait state. Not maximally
40  distinctive; typical.
41  - ONE verb / ONE beat in action. Don't stack multiple distinct actions.
42  - BANNED CUES -- distinctiveness markers. AVOID unless absolutely
43  natural: "immediately", "at once", "decisively", "deliberately",
44  "without hesitation", "instinctively", "swiftly", "instantly".
45  - When the response is a partial / silent / awkward reaction, write THAT.
46  - THE THOUGHT FIELD CARRIES PHASE VARIANCE. Adjacent phases may have nearly identical
47  action+speech but differ in HOW they interpret. Use this.
48  - gt_typicality:
49    "typical" -- modal behavior for this state distribution.
50    "plausible_tail" -- in distribution but not the mode. Use when the situation pulls slightly
51    off-phase, or when honoring the breadth of the state distribution is more honest than always
52    writing the cleanest exemplar.
53  - COMPACT-TIME ARCS: if multiple phases share the same life-stage, phase differences may live ONLY
54  in stance/thought, not in action-level changes.

```

TABLE 17 Shared style guideline (`_style_block`). Inlined verbatim into the user message of the three probe designers (Tables 22–24).

```

1 PHASE RESPONSES -- produce EXACTLY N entries, one per trajectory phase, in phase_idx order 0..N-1.
2
3 For each entry:
4   - phase_idx: 0..N-1 (matches the trajectory order)
5   - gt_action: <= 30 words (35 for In-Scenario). <character>'s action AT THIS PHASE, drawing ONLY
      on what this phase's character knows (events up to its query_chapter, NOT later events).
6   - gt_speech: <= 25 words spoken naturally, or null.
7   - gt_thought: 1-2 sentences. How THIS phase's character construes, interprets, or ruminates on
      the situation. THIS IS WHERE ADJACENT PHASES DIFFERENTIATE most when action is constrained.
8   - gt_typicality: "typical" (default) or "plausible_tail".
9
10 Phase list (each phase responds AS IF asked at its own query_chapter): phase_idx=0 "<label>" --
      <position_description (first 140 chars)> query_chapter=<q< (knows only chs 1..<q<)
      [life-stage info for Out-of-World]
11 ... phase_idx=N-1 ...
12
13 CRITICAL CONSTRAINTS:
14 - Adjacent phases that differ on the decision variable should differ in KIND (different verbs OR
      different thought-framings) -- not just intensity adverbs. If the scene's action affordance is
      narrow, put the variance in gt_thought.
15 - A phase that has not yet encountered a piece of information (because it comes from a later
      chapter) must NOT reference it.
16 - <character> stays in their established voice. No fourth-wall narration.

```

**TABLE 18** Shared phase-response instructions (`_phase_response_instructions`). Inlined verbatim into the user message of the three probe designers.

```

1  [System]
2  You are a personality construct analyst. Given a Character Arc -- a labeled trajectory of phases
   with chapter ranges and positions -- you name the SINGLE behavioral decision variable whose
   answer flips between phases. You think in terms of observable behavior and concrete choices,
   not abstract personality language.
3
4  [User]
5  [CHARACTER ARC CONTEXT block: novel, character, axis_name, dimension_label, pole_start, pole_end,
   trajectory phases.]
6
7  TASK:
8
9  1. Identify the SINGLE decision variable that captures what flips between phases. State it as a
   BINARY behavioral contrast that the character would be answering THROUGH ACTION. It must:
10     - be answerable through behavior/choice
11     - be specific enough that different phases give different answers
12     - be phrased neutrally (no preferred pole)
13     - be ONE dimension with two contrasting poles
14
15  2. For each adjacent phase pair, describe in one sentence how the decision-variable answer shifts.
   Produce EXACTLY (N-1) entries -- one per adjacent pair, in order -- with 0-indexed from/to
   indices.
16
17  Return JSON: {decision_variable, phase_contrasts: [{from_phase_idx, to_phase_idx, contrast}]}.

```

TABLE 19 Decision-variable extractor (Stage i of probe generation).

```

1  [System]
2  You are a literary-age analyst. Given a character's arc trajectory with chapter ranges and
   position descriptions, you assign each phase a developmental life-stage tag from a fixed
   taxonomy and a brief age estimate. You read chapter ranges and developmental cues in the
   position descriptions (school grade, marriage, retirement, parenting, etc.) rather than
   guessing.
3
4  [User]
5  [CHARACTER ARC CONTEXT block.]
6
7  TASK: For EACH phase, assign:
8  - life_stage: one of {child, adolescent, young_adult, adult, older_adult} child      =
   roughly 0-12 adolescent    = roughly 13-17 young_adult    = roughly 18-30 adult      =
   roughly 31-50 older_adult  = roughly 51+
9  - approx_age: a brief age estimate
10 - rationale: one sentence linking your tag to the phase's position_description and chapter range
11
12 PHASES:
13   Phase 0 (idx=0) "<label>" (chs. <a>-<b>):
14     <position_description>
15     ...
16
17   Produce EXACTLY N entries, in phase_idx order 0..N-1.
18
19   - If the source story spans many years, phases will progress through stages.
20   - If multiple phases share the same stage, that is fine -- assign honestly.
21   - For unusual chronologies (allegory, dream sequences), pick the stage that best fits the
     position_description rather than a literal age.
22
23   Return JSON: {phases: [{phase_idx, life_stage, approx_age, rationale}]}.

```

TABLE 20 Life-stage tagger.

|    |                                                                                                                                                                                                                                                                                                                    |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | [System]                                                                                                                                                                                                                                                                                                           |
| 2  | You are an abstract psychometric analyst. Given a Character Arc framed in a specific novel's setting, you re-express the arc in era-agnostic behavioral terms -- abstract psychological vocabulary that could apply to any time and place. You strip the era-specific scaffolding but preserve the trait dynamics. |
| 3  |                                                                                                                                                                                                                                                                                                                    |
| 4  | [User]                                                                                                                                                                                                                                                                                                             |
| 5  | [CHARACTER ARC CONTEXT block.]                                                                                                                                                                                                                                                                                     |
| 6  |                                                                                                                                                                                                                                                                                                                    |
| 7  | Re-express the Character Arc above in ERA-AGNOSTIC abstract psychological terms.                                                                                                                                                                                                                                   |
| 8  |                                                                                                                                                                                                                                                                                                                    |
| 9  | TASK:                                                                                                                                                                                                                                                                                                              |
| 10 | 1. abstract_axis: 1 sentence naming the abstract dimension (e.g.                                                                                                                                                                                                                                                   |
| 11 | "agency vs communion under pressure"). No novel-specific terminology.                                                                                                                                                                                                                                              |
| 12 |                                                                                                                                                                                                                                                                                                                    |
| 13 | 2. abstract_phases: EXACTLY N entries, one per trajectory phase in order. Each is a short phrase summarizing that phase's position on the abstract axis in era-agnostic terms.                                                                                                                                     |
| 14 |                                                                                                                                                                                                                                                                                                                    |
| 15 | Return JSON: {abstract_axis, abstract_phases: ["...", "..."]}                                                                                                                                                                                                                                                      |

TABLE 21 Era-agnostic axis re-expression (Out-of-World only).

```

1 [System]
2 You are a narrative psychologist building a behavioral probe from a verbatim source passage. You
  write ONE (scenario, question) plus one response per trajectory phase. The ANCHOR phase's
  response is extracted from the passage (what the character actually does there). Other phase
  responses are counterfactual projections -- "if this phase's character policy were applied to
  the same situation, what would they do and (more importantly) HOW would they process it?". The
  thought field carries most of the phase variance when action is constrained by the scene.
3
4 [User]
5 [CHARACTER ARC CONTEXT block.]
6
7 PROBE TYPE: (1) In-Scenario (multi-phase)
8 ANCHOR PHASE: idx=<k> "<label>" (chs <a>-<b>)
9 ANCHOR QUERY CHAPTER: <qc>
10 [Relational arcs only: RELATIONAL TARGET block stating that every phase response describes
   <character>'s behavior directed at <target>.]
11
12 SOURCE PASSAGE (verbatim -- contains the ANCHOR phase's canonical response):
13 ""
14 <verbatim_passage>
15 ""
16
17 TASK: Assemble ONE probe from this passage with N phase responses.
18
19 1. SCENARIO: paraphrase the SETUP from the passage -- location, time, present characters, leading
   up to <character>'s response. Stop just before <character> acts/speaks.
20
21 2. QUESTION: name the specific choice point (see QUESTION STYLE).
22
23 3. <Phase-response instructions (shared sub-block).>
24
25 ANCHOR phase entry (phase_idx=<k>) MUST extract <character>'s actual response from the SOURCE
   PASSAGE -- gt_action paraphrases the passage; gt_speech copies <character>'s spoken words
   verbatim from the passage (or null if they don't speak); gt_thought captures the construal
   evident in the passage. Non-anchor phases are counterfactual projections.
26
27 <Style guidelines (shared sub-block).>
28
29 NOTE for (1) In-Scenario: anchor scenes often have narrow behavioural affordance -- late-phase
   trait policies may legitimately produce similar actions to early phases in cramped scenes.
   When that happens, push the differentiation into gt_thought (cognitive construal differs even
   when action is similar). Do not stretch action variance to manufacture differentiation.
30
31 Return JSON: {scenario, question,
32               phase_responses: [{phase_idx, gt_action, gt_speech, gt_thought, gt_typicality}]}.
```

**TABLE 22** In-Scenario designer (probe type 1). The user message also inlines the shared phase-response instructions (Table 18) and style guidelines (Table 17).

```

1 [System]
2 You are a narrative psychologist designing a multi-phase behavioral probe in the source novel's
   world. The scenario is a plausible-but- unwritten situation in the source world, NEVER
   reproducing any specific canonical scene. You write ONE (scenario, question) plus one response
   per trajectory phase. Each phase response is the typical behavior of THAT phase's character --
   action, optional speech, and explicit thought process (how this phase construes the
   situation). You preserve the character's established voice, time period, and world rules.
   Adjacent phases may legitimately respond similarly; do not stretch.
3
4 [User]
5 [CHARACTER ARC CONTEXT block.]
6
7 PROBE TYPE: (2) In-World (multi-phase)
8 ANCHOR PHASE: idx=<k> "<label>" (chs <a>-<b>)
9 ANCHOR QUERY CHAPTER: <q>
10 ANCHOR PHASE POSITION: <position_description>
11 [Relational arcs only: RELATIONAL TARGET block stating that the scenario MUST involve direct
    interaction with <target>.]
12
13 HARD CONSTRAINTS:
14 - The scenario must NOT reproduce any specific canonical scene from the source novel. It must be a
    plausible-but-unwritten moment in the source world.
15 - <character> must remain in their established voice and time period.
16 - The situation must be PHASE-AGNOSTIC enough that any of the trajectory's phases could plausibly
    face it (don't bind to a single phase's specific life-situation).
17 - The situation must MEANINGFULLY ACTIVATE the decision variable so different phases of
    <character> would respond differently.
18
19 TASK:
20
21 1. SCENARIO: a plausible-but-unwritten situation around chapter <q>. End just before <character>
    responds. Keep it phase-agnostic.
22
23 2. QUESTION: pose the choice point (see QUESTION STYLE).
24
25 3. <Phase-response instructions (shared sub-block)>.
26
27 <Style guidelines (shared sub-block)>.
28
29 Return JSON: {scenario, question, phase_responses: [...]}.

```

**TABLE 23** In-World designer (probe type 2). The user message also inlines the shared phase-response instructions (Table 18) and style guidelines (Table 17).



```

1 [System]
2 You are a narrative psychologist transposing a character's narrative identity into a NON-source
  era. You receive an explicit target era and a per-phase life-stage table. The scenario is set
  in the target era and is AGE-AGNOSTIC -- described so that the same situation can be responded
  to by the character at multiple life-stages. You write ONE (scenario, question) plus one
  response per trajectory phase, with each phase response embodying ITS OWN life-stage (a child
  phase's response describes child-Harry; an adult phase's response describes adult-Harry, same
  era). The PROTAGONIST is the named character throughout; identity stays constant, props/age
  vary by phase. No source-world props (no magic, no Hogwarts, no Regency drawing rooms, no
  slave ships) ever.
3
4 [User]
5 [CHARACTER ARC CONTEXT block (with life-stages appended per phase).]
6
7 PROBE TYPE: (3) Out-of-World (multi-phase, non-source era)
8 ANCHOR PHASE: idx=<k> "<label>"
9 ABSTRACT AXIS: <abstract_axis>          # from Table
10                                     # ref{tab:prompt-axis-re-exp}
11
12 ABSTRACT PHASE POSITIONS (era-agnostic): phase_idx=0: <abstract_phase_0>
13 ...
14
15 TARGET ERA (shared across all phase responses): <era_label>
16   Description: <era_description>
17 [Relational arcs only: RELATIONAL TARGET block stating that <target> also appears with the same
  name but a role appropriate to the target era.]
18
19 LIFE-STAGE LOCK -- each phase response describes <character> at THAT phase's life-stage in the
  target era. The scenario MUST be age-agnostic -- a situation that could plausibly happen to
  <character> at any of the listed life-stages (with different stakes and capabilities, but
  recognizably the same situation).
20
21 PROTAGONIST LOCK -- STRICT:
22 - The protagonist is <character> (same name, same core identity).
23 - The response is about <character> in every phase entry.
24 - Never re-cast under another name.
25
26 HARD CONSTRAINTS:
27 - Target era ONLY: vocabulary, technology, institutions, idioms fit the TARGET era for every phase
  response.
28 - NO source-world props (no magic, no Hogwarts, no Regency drawing rooms, no slave ships, etc.)
  ever.
29 - The situation must activate the abstract axis so different phases' trait states have
  recognizably different behavioral expressions.
30
31 TASK:
32
33 1. SCENARIO: a brief age-agnostic situation in the target era centered on <character>. Set the
  tension in 1-2 sentences (<=60 words). End just before <character> responds.
34
35 2. QUESTION: pose the choice point (see QUESTION STYLE).
36
37 3. <Phase-response instructions (shared sub-block).>
38
39 <Style guidelines (shared sub-block).>
40
41 Return JSON: {scenario, question, phase_responses: [...]}.

```

**TABLE 24 • Out-of-World designer (probe type 3).** The user message also inlines the shared phase-response instructions (Table 18) and style guidelines (Table 17).

```

1  [Locator -- System]
2  You are an editorial archivist for literary text. Given a brief moment description and a list of
   candidate events from the same chapter range, you select the one event whose content most
   directly enacts the moment. You reject the entire list if none enacts the moment.
3
4  [Locator -- User]
5  KEY MOMENT to locate:
6  "<key_moment>"
7
8  CHAPTER RANGE for this arc phase: <a>-<b>
9
10 CANDIDATE EVENTS:
11   Chapter  <c> . <event_id>: <event_description>
12   ...
13
14 TASK: Pick the ONE event whose CONTENT most directly enacts the key moment.
15 - Match by event semantics, not keyword overlap.
16 - For late-arc transformation moments, prefer candidates near the END of the chapter range.
17 - If no candidate genuinely depicts the moment, set event_id to null.
18
19 Return JSON: {chapter, event_id, reasoning}.
20
21 [Extractor -- System]
22 You are an editorial archivist for literary text. Given a chapter and a target moment, you locate
   the exact passage and return it verbatim -- character-for-character, no paraphrase, no
   compression, no invention. You always include enough surrounding text to establish the scene
   AND show the character's actual response.
23
24 [Extractor -- User]
25 Below is Chapter <chapter_num> of "<novel>".
26
27 You are looking for the specific passage that depicts this moment: Key moment    : "<key_moment>"
   Event detail : "<event_description>"
28
29 TASK: Find the passage and return:
30 - verbatim_passage: EXACT source text, character-for-character. Include enough text to establish
   the scene AND show the character's response. Typically 2-5 paragraphs (200-1500 words).
31 - first_words / last_words: opening and closing 5-8 words.
32
33 Do NOT paraphrase. Do NOT invent. If the moment is absent from this chapter, set verbatim_passage
   to null and explain in "note".
34
35 CHAPTER TEXT:
36 <chapter_text>
37
38 Return JSON: {verbatim_passage, first_words, last_words, note}.

```

**TABLE 25** In-Scenario scene grounding (locator + extractor). Two calls, executed sequentially per (axis, anchor phase).

```

1 [System]
2 You are a literary editor judging whether a single phase response sounds like that character at
   THAT phase, in the given setting. You catch anachronistic vocabulary for the setting (e.g.,
   'group chat' in Regency drawing-room; 'pull camera times' in 1990s magical Britain),
   broken-character self-narration, generic dialogue, and tonal mismatches. You judge action,
   speech, AND thought together -- the thought field must use language and concepts plausible for
   this character at this phase. gt_speech being null is NOT a failure on its own.
3
4 [User]
5 CHARACTER: <character>
6 PHASE: idx=<k> "<label>" -- <position_description>
7 KNOWLEDGE CUTOFF: phase responds as if asked at chapter <q> (knows only events up to that
   chapter).
8 SETTING: <"target era 'X'", or "source novel's world at chs a-b", depending on probe type>.
9
10 PROBE SCENARIO:
11   <scenario>
12
13 THIS PHASE'S RESPONSE:
14   Action:   <gt_action>
15   Speech:   <gt_speech or "(none -- pure action response)">
16   Thought:  <gt_thought>
17   Typicality: <gt_typicality>
18
19 TASK: Does this response sound like <character> at this phase, in this setting, given only
   knowledge up to chapter <q>? Catch:
20 - anachronistic word choice for the SETTING
21 - references to events the phase cannot know yet
22 - broken-character self-narration
23 - generic dialogue or thought
24 - tonal mismatches (young character with adult articulation; vice versa)
25
26 Note: gt_speech being null is NOT a failure.
27
28 - verdict: "pass" if response is in character; "fail" otherwise.
29 - note:    1 sentence -- what's right or wrong.
30
31 Return JSON: {verdict, note}.

```

TABLE 26 Q-Voice validator (per response).

```

1 [System]
2 You are a psychometric reviewer. Given a Character Arc's trajectory (each phase's
   position_description), the decision variable, and ONE phase response (action + speech +
   thought), you identify the trajectory phase this response is most diagnostic of. You judge
   from the WHOLE response -- action, speech, and especially thought (how the character construes
   the situation). You are NOT told which phase generated this response. Pick the phase whose
   POSITION DESCRIPTION best matches what the response actually expresses, not the phase whose
   label sounds closest.
3
4 [User]
5 DECISION VARIABLE: <decision_variable>
6 POLE START:      <pole_start>
7 POLE END:        <pole_end>
8
9 ARC PHASES (0..N-1): idx=0 "<label>" -- <position_description>
10 ...
11
12 PROBE:
13   Scenario:      <scenario>
14   Question:      <question> Response action: <gt_action> Response speech: <gt_speech or
   "(none)"> Response thought: <gt_thought>
15
16 TASK: Based on the response's ACTION + SPEECH + THOUGHT, identify which trajectory phase this
   response is most diagnostic of.
17
18 - most_likely_phase_idx: integer 0..N-1. Pick the phase whose POSITION DESCRIPTION best matches
   what the response actually expresses. Use the THOUGHT field heavily -- it often carries
   phase-distinguishing cognitive style when action is constrained.
19 - confidence: "low" | "medium" | "high".
20 - note:      1 sentence -- the deciding cue.
21
22 You are NOT told which phase generated this probe. Judge solely on what the response expresses.
   Adjacent classification is fine and expected (phases legitimately overlap). If two adjacent
   phases fit equally well, pick either and use confidence="medium" or "low".
23
24 Return JSON: {most_likely_phase_idx, confidence, note}.
25
26 Wrapper post-processes to a verdict:
27   pass      -- most_likely_phase_idx == target_phase_idx adjacent -- |most_likely - target| <= 1
   off_phase-- |most_likely - target| >= 2

```

TABLE 27 Q-PhaseFit validator (per response, blind to target phase).

```
1 [System]
2 You are an editorial fact-checker. Given a source verbatim passage and a constructed (scenario,
   question, anchor-phase response) triplet derived from it, you verify that the scenario matches
   the passage setup, the anchor-phase response action+speech matches the character's actual
   response in the passage, and nothing was invented.
3
4 [User]
5 SOURCE PASSAGE (verbatim):
6 ""
7 <source_passage>
8 ""
9
10 CONSTRUCTED PROBE:
11   Scenario: <scenario>
12   Question: <question> Anchor phase (idx=<k>) response:
13     Action: <gt_action>
14     Speech: <gt_speech or "(none)">
15     Thought: <gt_thought>
16
17 TASK: Verify scenario + anchor-phase response are faithful to the source.
18 - Scenario reflects the passage's setup (location, present characters, tension)?
19 - Anchor action+speech matches what the character actually does in the passage (paraphrase OK;
   invention NOT OK)?
20 - Thought is plausibly inferable from what the passage shows of the character's mind in this beat?
21
22 - verdict: "pass" / "fail"
23 - note: 1 sentence.
24
25 Return JSON: {verdict, note}.
```

TABLE 28 Q-Anchor validator (per probe; In-Scenario only).

```

1 [System]
2 You are a worldbuilding reviewer. You verify that a probe's scenario and phase responses respect
   the rules of the probe's setting. For In-World probes you check source-world rules (era,
   technology, magic) and that no canonical scene is reproduced. For Out-of-World probes you
   check internal consistency within the named target era, the absence of any source-world props,
   and that each phase response describes the character at the life-stage assigned to that phase.
3
4 [User]
5 PROBE TYPE CONTEXT:
6 [For Out-of-World:]
7 (3) Out-of-World -- the scenario is set in TARGET ERA "<era_label>".
8   Description: <era_description> Per-phase life-stage requirements: phase_idx=0: must be
   <life_stage> (~<approx_age>)
9   ...
10  CHECK:
11    (a) target-era internal consistency (no anachronisms);
12    (b) NO source-world props leak in (no magic, no Hogwarts, etc.);
13    (c) each phase response describes <character> at its required life-stage;
14    (d) <character> is the named protagonist throughout.
15
16 [For In-World:]
17 (2) In-World -- the scenario is in the world of "<novel>" around chapter <anchor_query_chapter>.
18  CHECK:
19    (a) scenario obeys the source world's rules (era, technology, magic system if any);
20    (b) only characters who would exist at this point in the story are present;
21    (c) nothing reproduces a specific canonical scene from the source.
22
23  CHARACTER: <character>
24  SCENARIO: <scenario>
25
26  PHASE RESPONSE SUMMARIES:
27    phase_idx=0: action=<truncated> | thought=<truncated>
28    ...
29
30  TASK:
31  - verdict: "pass" if the rules above are respected throughout,
32            "fail" otherwise.
33  - note: 1 sentence -- what's consistent or inconsistent.
34
35  Return JSON: {verdict, note}.

```

TABLE 29 Q-World validator (per probe; In-World and Out-of-World).

```

1  [System]
2  You are a psychometric reviewer judging cross-phase variation within one probe. Given the decision
   variable, the trajectory's phase descriptions, and N phase responses to the SAME
   scenario+question, you judge whether each adjacent pair is differentiated on the
   decision-variable axis. The differentiation can live in ACTION, SPEECH, or THOUGHT -- when
   scenes constrain action, the cognitive style often carries the phase signal. Adjacent overlap
   is theoretically expected when position descriptions are themselves close -- this is
   reporting, not gatekeeping.
3
4  [User]
5  DECISION VARIABLE: <decision_variable>
6
7  SHARED SCENARIO: <scenario>
8  SHARED QUESTION: <question>
9
10 PHASE RESPONSES (all to the SAME scenario+question):
11
12 Phase idx=0 "<label>" (pos: <position_description (first 100)>...)
13   Action: <gt_action>
14   Speech: <gt_speech or "(none)">
15   Thought: <gt_thought>
16
17 ... (one block per phase)
18
19 TASK: For each ADJACENT phase pair, judge whether the two responses are differentiated on the
   decision variable. The differentiation can live in ACTION, SPEECH, or (often, when the scene
   constrains action) THOUGHT -- phase-specific cognitive construal carries the trait signal when
   behavior is similar.
20
21 For each adjacent pair (idx k, idx k+1) of the trajectory:
22   - separation:
23     "separated" -- responses commit to genuinely different decision-variable answers, in the
       direction phases imply.
24     "similar" -- responses give the same/near-same answer; this is FINE if the phases'
       position_descriptions themselves are close.
25     "ambiguous" -- cannot tell because responses cite different stakes or framings.
26   - note: 1 sentence -- what cued your judgment.
27
28 Return weak_pairs[] containing entries for pairs where separation is
29 "similar" or "ambiguous". If a pair is "separated", omit it from weak_pairs.
30
31 This is annotation, NOT gatekeeping -- nothing drops on this verdict.
32
33 Return JSON: {weak_pairs: [{from_phase_idx, to_phase_idx, separation, note}], note}.

```

TABLE 30 Q-Discrim validator (per probe; annotation-only, never drops).



|    |                                                                                                 |
|----|-------------------------------------------------------------------------------------------------|
| 1  | [System]                                                                                        |
| 2  | You are <character>, from "<novel>". You are at the point in the story corresponding to chapter |
| 3  | <query_chapter>.                                                                                |
| 4  | [If the axis is relational:]                                                                    |
| 5  | You are interacting with <target_character>.                                                    |
| 6  |                                                                                                 |
| 7  | [If context is non-empty:]                                                                      |
| 8  | Background you have access to:                                                                  |
| 9  | <context> <mode-specific context block; see Table ref{tab:prompt-per-mode-context}> </context>  |
| 10 |                                                                                                 |
| 11 | [User]                                                                                          |
| 12 | Scenario:                                                                                       |
| 13 | <scenario>                                                                                      |
| 14 |                                                                                                 |
| 15 | Question:                                                                                       |
| 16 | <question>                                                                                      |
| 17 |                                                                                                 |
| 18 | [For TimeCHARA only, when a hint is produced:]                                                  |
| 19 | (HINT: <hint>)                                                                                  |

**TABLE 31** Shared role-play system + user templates (all six modes use these; modes differ only in what is inserted into the `<context>` block, see Table 32).

```

1  [Vanilla]
2  (empty -- no context block is added)
3
4  [Summary]
5  <concatenated chapter summaries for chapters query_chapter-4 .. query_chapter (i.e.
6    SUMMARY_MAX_CHAPTERS=5)>
7
8  [RAG]
9  <top-6 source-text chunks retrieved with (scenario + question) as the embedding query, filtered to
10   chunks at chapter <= query_chapter; 1500-char chunks with 300-char overlap;
11   text-embedding-3-small>
12
13 [LifeChoice]
14 [Character Description]
15 <concatenated chapter summaries, same as Summary mode>
16
17 [Relevant Memory]
18 [Chapter <c> excerpt]
19 <chunk text>
20 ...
21 [Chapter <c> excerpt]
22 <chunk text> (top-6 retrieved chunks; the embedding query is the Character Description instead of
23   the (scenario + question) pair)
24
25 [TimeCHARA]
26 (empty -- TimeCHARA injects information via the user-side hint, not via the context block.)
27
28 [Arc (ours)]
29 <final-stage axis JSON for the relevant axis, with all phases at chapter index > query_chapter
30   hidden; `literary_validation` and `evidence_summary` always stripped; when at least one later
31   phase is hidden, `pole_end` and `arc_direction` also stripped>

```

**TABLE 32** Per-mode `<context>` payloads. TimeCHARA’s two-stage hint pipeline is given separately in Table 33.

```

1  [Shared expert system]
2  You are a helpful and accurate assistant.
3
4  [Temporal expert -- user]
5  You will be given a scenario and a question from the novel "<novel>". The novel has
   <total_chapters> chapters. Your task is to identify which chapter contains the scene the
   scenario describes.
6  ***
7  [Scenario]
8  <scenario>
9  [Question]
10 <question>
11 ***
12 [Steps]
13 1. Recall the scene from the scenario and describe it using the six Ws (Who, What, When, Where,
   Why, How).
14 2. Identify the single chapter number (1-<total_chapters>) that contains this scene. If you cannot
   determine it, output "unknown".
15
16 First, reason step by step. Then on its own final line, output ONLY
17 "Chapter: N" (an integer) or "Chapter: unknown".
18
19 [Spatial expert -- user]
20 You will be given a scenario, a question, and a character from "<novel>". Your task is to classify
   whether the character is present in the scene described.
21 ***
22 [Scenario]
23 <scenario>
24 [Question]
25 <question>
26 [Character]
27 <character>
28 ***
29 [Steps]
30 1. Recall the scene and list every character involved, including those present but unmentioned.
31 2. Decide whether <character> is among them.
32
33 First, reason step by step. Then on its own final line, output ONLY
34 "Presence: present" or "Presence: absent".
35
36 [Future-event hint string]
37 Note that the period of the question is in the future relative to <character>'s time point.
   Therefore, you should not answer the question or mention any facts that occurred after
   <character>'s time point.
38
39 [Absent-character hint string]
40 Note that <character> had not participated in the scene described in the question. Therefore, you
   should not imply that <character> was present in the scene.

```

**TABLE33** TimeCHARA two-stage hint pipeline. The spatial-expert call is skipped when the temporal expert returns a chapter `> query_chapter`; in that case the future-event hint is appended directly.

```
1 [System]
2 You are a literary summarizer. Given one chapter of a novel, you write a concise prose summary
   covering the events, who was present, what was said or decided, and the emotional shape of the
   chapter. You stay faithful to the text and do not invent details.
3
4 [User]
5 Novel: <novel>
6 Chapter: <chapter_idx>
7
8 --- chapter text ---
9 <chapter_text>
10 --- end ---
11
12 Write the summary as one or two paragraphs of running prose.
```

TABLE 34 Chapter summarizer (used by Summary and LifeChoice).

```

1 [System]
2 You are an evaluator of character-grounded narrative responses. Compare a model's response to the
  Reference and score it on three dimensions, each from 1 to __SCALE__, using a phase-fidelity /
  mechanism framework.
3
4 The Reference represents how this character behaves at the target arc phase. Score whether the
  response is mechanism-equivalent to the *same* phase, not whether it merely looks like the
  character "in general."
5
6 Principles:
7 - The reference is an anchor, not a surface form. Score functional / mechanistic equivalence, not
  word-matching.
8 - Respect negative space. If the reference shows silence or inaction, treat that absence as part
  of the reference.
9 - Beware stereotype regression. When the reference shows the character resisting their canonical
  template (heroic, mentorly, etc.), penalize responses that fall back to the template.
10 - Penalize fabrication. Reference-absent props, names, or backstory that drive the response should
  lower the score.
11 - Respect the character's epistemic state. No future-event awareness, no information the character
  couldn't have at this point.
12
13 Dimensions:
14
15 1. APF -- Action Phase-Fidelity. Is the response's action mechanism- equivalent to ref_action at
  the target phase? Judge across three levels (Level A is strictest; B and C can partially
  salvage credit when A fails):
16 - Level A -- Strategy match: same underlying strategy (e.g. withdrawal, confrontation,
  deflection, mediation, concealment, disclosure, support-seeking).
17 - Level B -- Valence match: same emotional / relational valence (positive engagement / negative
  withdrawal / neutral observation).
18 - Level C -- Target match: same target (same person, environment, or internal state). A match
  on all three is high; matching only B and C is mid; missing all three cannot be high. The
  top-level apf score should weight Level A heaviest, since strategy is what phase-fidelity
  ultimately rides on.
19
20 2. RPF -- Reasoning Phase-Fidelity. Parse both ref_thought and the response's reasoning into four
  mechanism slots and judge each slot:
21 - Trigger: what event / utterance / internal state set this response in motion?
22 - Appraisal: how does the character interpret / evaluate that trigger?
23 - Goal: what short-term goal arises from the appraisal?
24 - Strategy: how is the goal executed? Slot weights are NOT equal. The top-level rpf score
  should weight Strategy and Appraisal heaviest -- that is where phase- misalignment surfaces
  first. Trigger and Goal contribute less. If the response does not externalize its reasoning,
  infer the mechanism from the action.
25
26 3. RAE -- Reasoning-Action Entailment. Given the *reference reasoning (ref_thought)* as a fixed
  anchor, judge whether the response's action is a plausible action that this reasoning would
  license at this phase. This dimension is *conditioned on ref_thought*, not on the response's
  own reasoning -- a response can have internally self-consistent reasoning that still fails to
  follow from ref_thought, and that is the failure mode RAE is designed to catch. Three
  sub-checks:
27 - gt_entailment: Treating ref_thought as the operative reasoning, is the response's action
  among the plausible actions that ref_thought would license? An action that contradicts
  ref_thought's appraisal, goal, or strategy fails here even if the action looks reasonable in
  isolation.
28 - phase_consistency: Do the response's action and the response's reasoning belong to the same
  arc phase? (Phase-mismatch
29 failure: action looks like one phase, reasoning like another -- even when each half is
  independently plausible.)
30 - direction_consistency: Are the response's action and reasoning directionally aligned with
  each other? (Internal contradiction: reasoning "avoid the threat" but action "attack the
  threat".) The top-level rae score should reflect that any single low sub- check is enough to
  call the (action, reasoning) pair inadequately entailed. gt_entailment is the dominant
  sub-check -- the entire dimension exists to verify that the action follows from ref_thought,
  not merely that the response is internally coherent.

```

```

1 [System]
2 You are an evaluator of character-grounded narrative trajectories. Compare a model's *sequence of
  responses across multiple arc phases* to the *sequence of references (reasoning + action)
  across the same phases*, and score how faithfully the model reproduces the trajectory of
  change. Score on three sub-checks (Phase Trajectory Fidelity, PTF), each from 1 to __SCALE__.
3
4 Each Reference at phase i represents how this character behaves at that arc phase. Each model
  response at phase i is the model's attempt to behave as this character at the same phase. Your
  task is to look across all phases together and judge whether the model's trajectory of change
  mirrors the reference trajectory of change -- not whether any single phase response is good in
  isolation.
5
6 Input format:
7 You will receive SCENARIO, QUESTION, and then N phase blocks in chronological order. Each block is
  headed by `[PHASE <phase_idx>]` and contains the reference fields (ref_action / ref_speech /
  ref_thought) and the model's response at that phase (model_response). Phase indices are taken
  from the source probe and may have gaps (an
8 "unavailable" reference phase is omitted) -- use the indices as given.
9
10 Principles:
11 - Look across phases, not within one. A response that is locally fine at phase i can still belong
    to the wrong phase in the trajectory; that is the failure mode this evaluation is designed to
    catch.
12 - Reward differentiation only when it is grounded. A model that produces different responses
    across phases scores well only if those differences track the reference's differences --
    random or stylistic variation does not count.
13 - Penalize phase collapse. A model that gives near-identical responses across phases has failed
    regardless of how well any single response matches its own phase.
14 - Penalize phase swapping. A response that better fits another phase's reference than its own is a
    trajectory error.
15 - Penalize fabrication and stereotype regression. Reference-absent props/backstory, or a fallback
    to the character's canonical template against a reference that resists it, should pull scores
    down.
16 - Respect the character's epistemic state at each phase -- no future- event awareness, no
    later-phase information leaking back into earlier-phase responses.
17 - An empty / refused response at a phase ("[empty response -- model returned nothing]" or similar)
    is part of the trajectory: treat it as a failed anchor at that phase and as a missing
    contribution to direction/shape.
18
19 Sub-checks:
20
21 1. ptf_alignment -- Per-phase anchoring. For each phase i, is the model's response at phase i
    closer in mechanism to the reference at phase i than to references at other phases? A high
    score means most responses are correctly anchored to their own phase; a low score means
    responses systematically drift toward the wrong phase or collapse into one undifferentiated
    voice.
22
23 2. ptf_direction -- Direction of change. Treating the reference sequence as defining a direction
    (e.g. fearful -> confident, dependent -> autonomous), does the model's sequence move along the
    same axis and in the same direction? A high score means the model's overall
    direction-of-travel matches the reference's; a low score means the model moves on a different
    axis, in the opposite direction, or shows no movement.
24
25 3. ptf_shape -- Shape and pacing of change. Beyond direction, does the model reproduce where on
    the trajectory the largest shifts occur and how gradual or abrupt those shifts are? A high
    score means inflection points and pacing line up with the reference; a low score means the
    model's trajectory has a clearly different internal structure (e.g. linear ramp where
    reference shows a sharp turning point, or inflection at the wrong phase). When N=2 there is no
    inflection structure to compare -- score shape on whether the magnitude of the single
    transition is similar to the reference's.
26
27 Output Format:
28 You score one (character, scenario) item across N phases. Return a single JSON object with this
    schema:
29

```